Heterogeneous RBCs via Deep Multi-Agent Reinforcement Learning

unifying macroeconomic modelling through computation

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*All views and opinions are those of the speaker(s) and do not necessarily reflect the position of Bank of Italy

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EUROSISTEMA

Acknowledgments first!

Joint work with

Federico **Gabriele**
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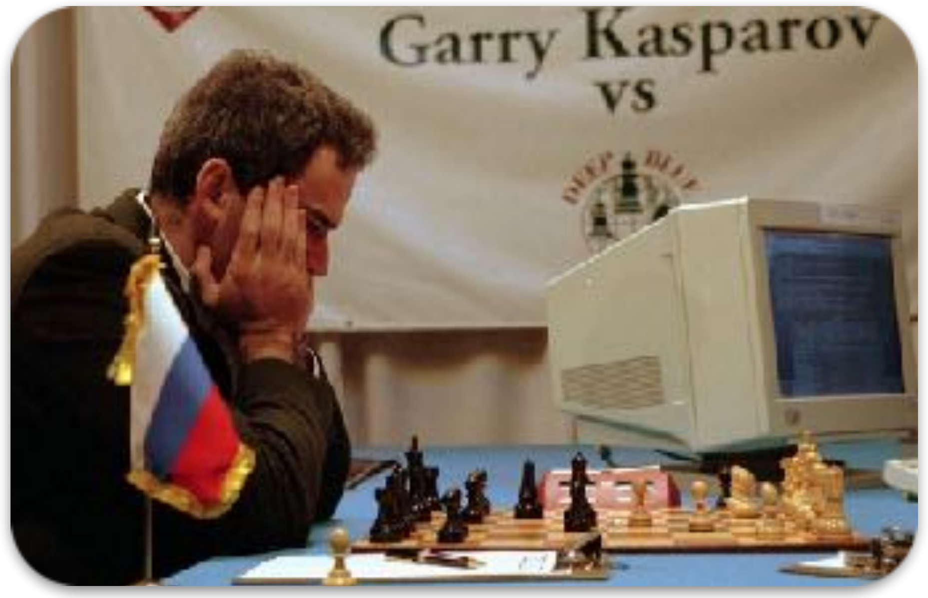
Marco **Taboga**
REI@BdI

we also thank

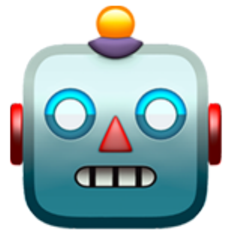
Alessandro Moro, Valerio Astuti, Valerio Nispi Landi, Sergio Santoro (Banca d'Italia), Doyne Farmer (Oxford University), Sara Casella (LUISS University)

Modelling and computation



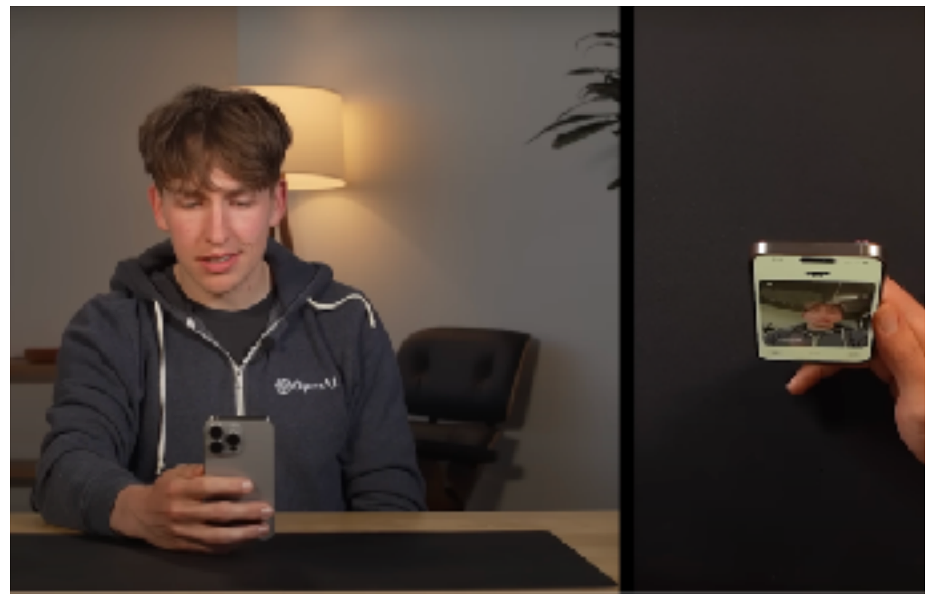
Traditional models

Based on sophisticated human-designed carefully crafted rules and databases



ML-driven models

Based on algorithms that can *self-improve* using *more data* and/or *more computation*



How can we enable this transition in econ?

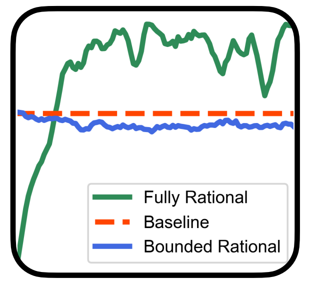
GitHub
bancaditalia

Line **1.** High-quality **open-source** models

Line **2.** Faster ML-based **calibration**

today's paper

Line **3.** **AI-driven** economic agents



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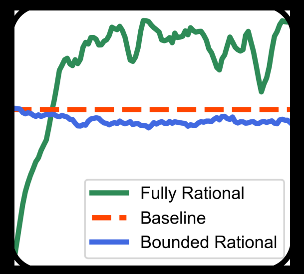
Line 3 **AI-driven economic agents**

with A. **Agrawal**, T. **Bacaloni**, S. **Brusatin**, A. **Coletta**, J. **Dyer**, D. **Delli Gatti**, F. **Gabriele**, T. **Padoan**, M. **Taboga**, M. **Wooldridge**



Simulating the Economic Impact of Rationality through Reinforcement Learning and Agent-Based Modelling **ICAIF (2024)**

Robust Policy Design in Agent-Based Simulators using Adversarial Reinforcement Learning, **MARW-AAAI (2025)**

Natural-gas storage modelling by deep reinforcement learning, **ICAIF (2025)**

Heterogeneous RBCs via Deep Multi-Agent Reinforcement Learning, **AAMAS (2026)**

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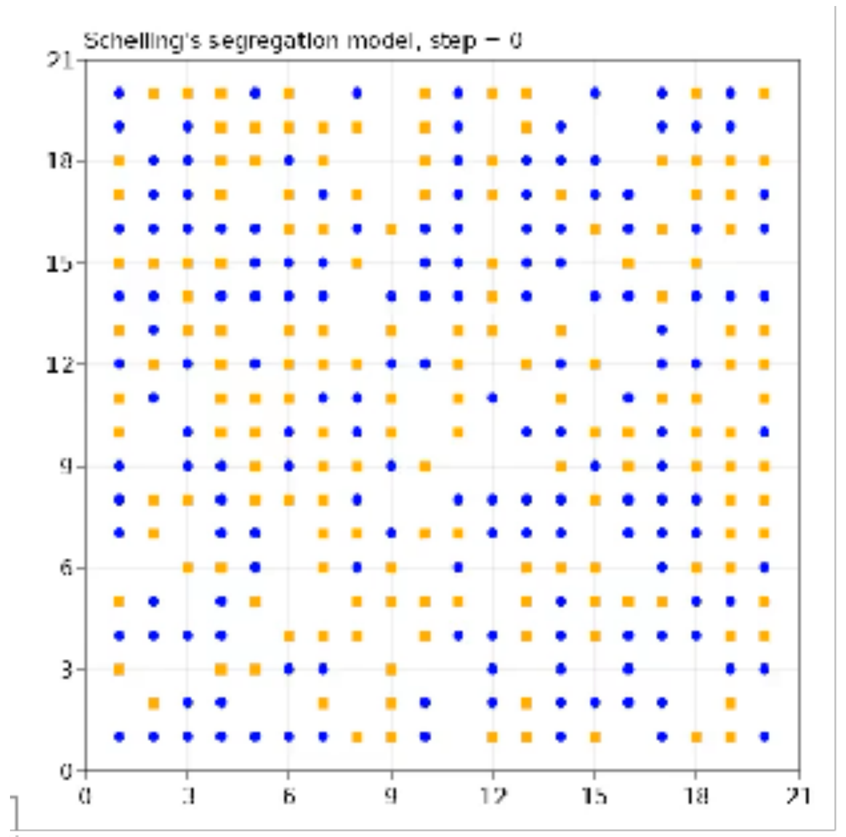
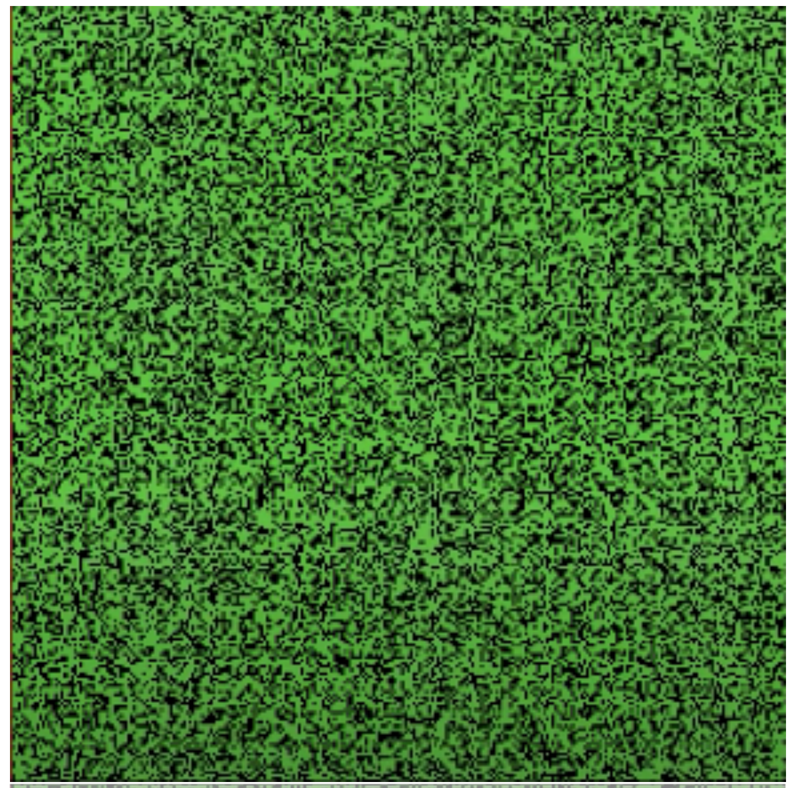
ABMs in a nutshell

Agent-based models (ABMs) grew from toy models to realistic simulations with **thousands of heterogeneous interacting** agents

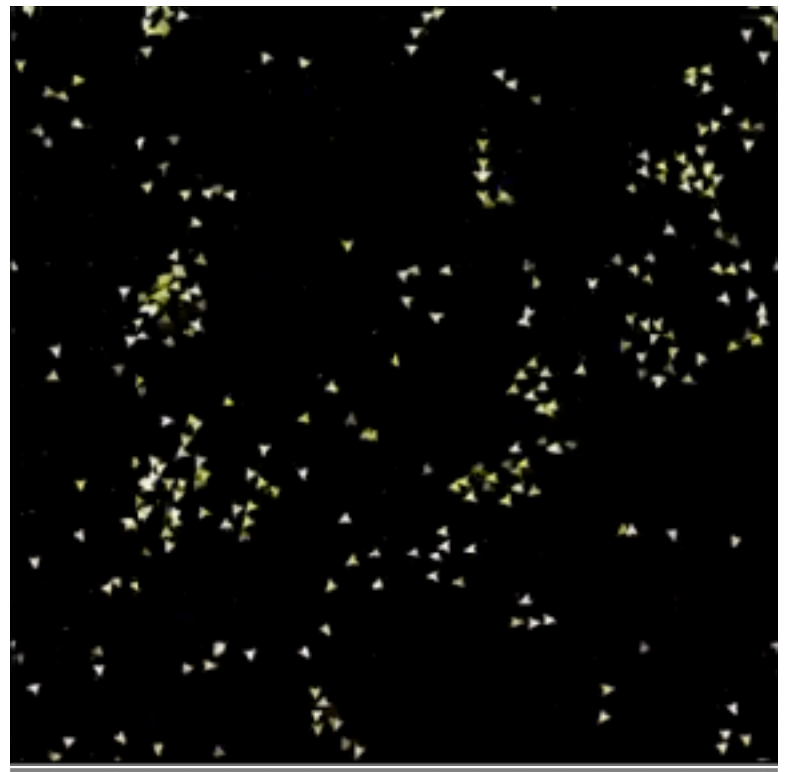
1980's — 2000's — 2020's



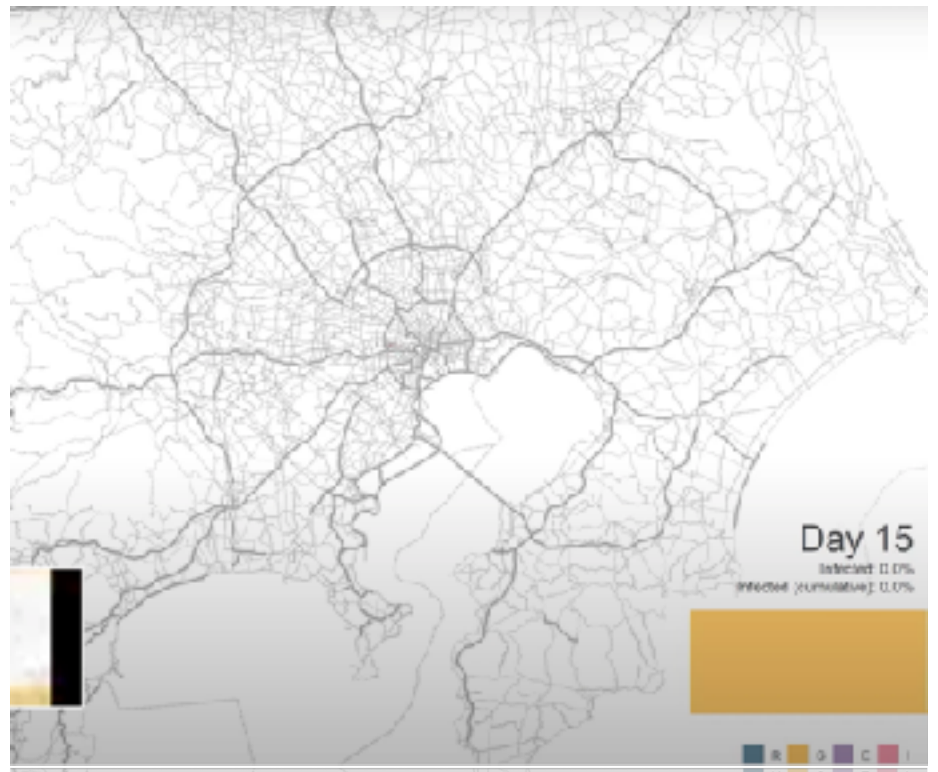
Shelling's segregation



Forest fires



Flocking



Large scale epidemic spreading

ABMs in a nutshell

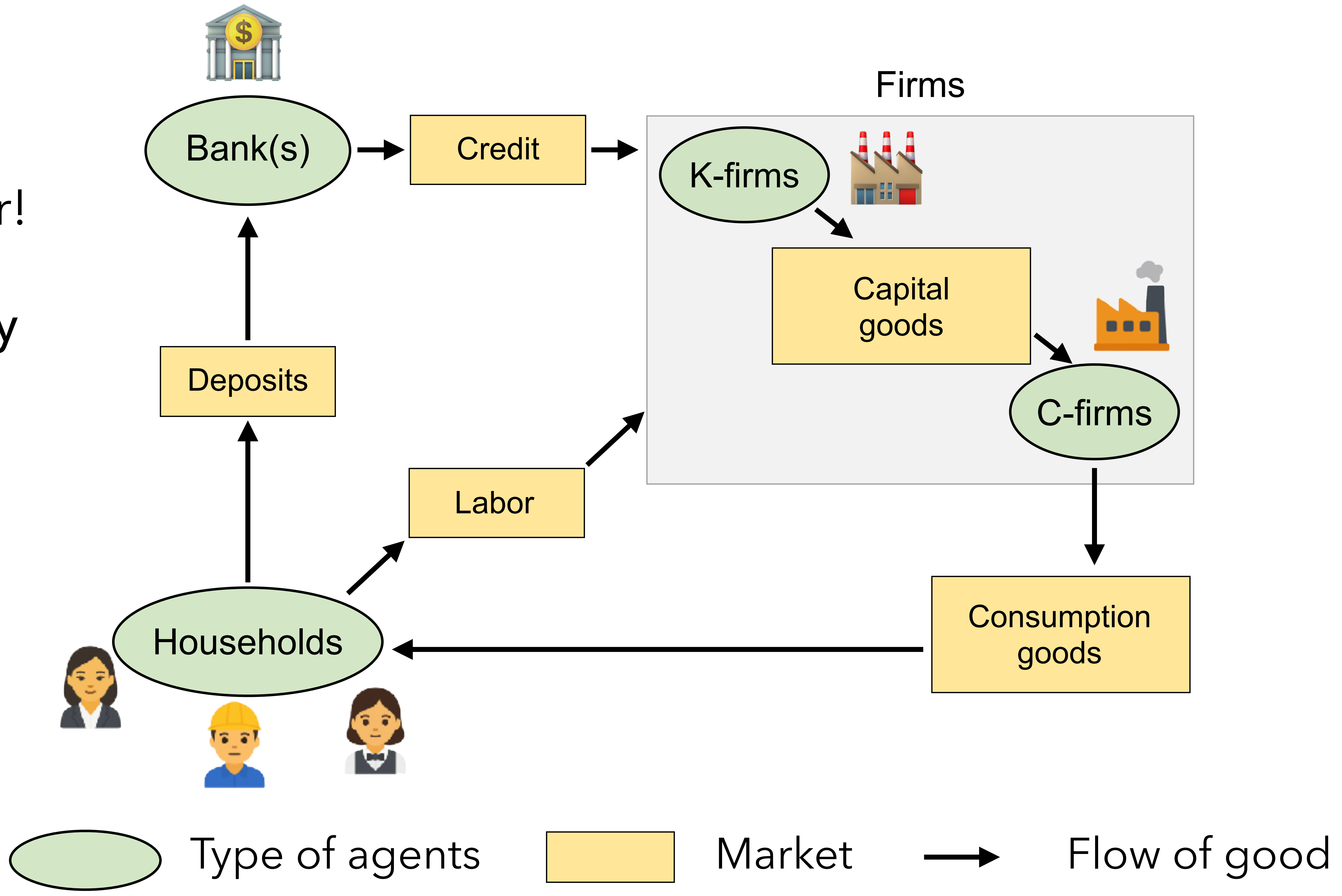
✔ Unconstrained **heterogeneity**!

✔ **Realistic**, boundedly rational behaviour!

✘ Behavioural **rules are difficult to specify** in advance…

✘ Behaviour does not necessarily adjust as a result of a **changing policy**

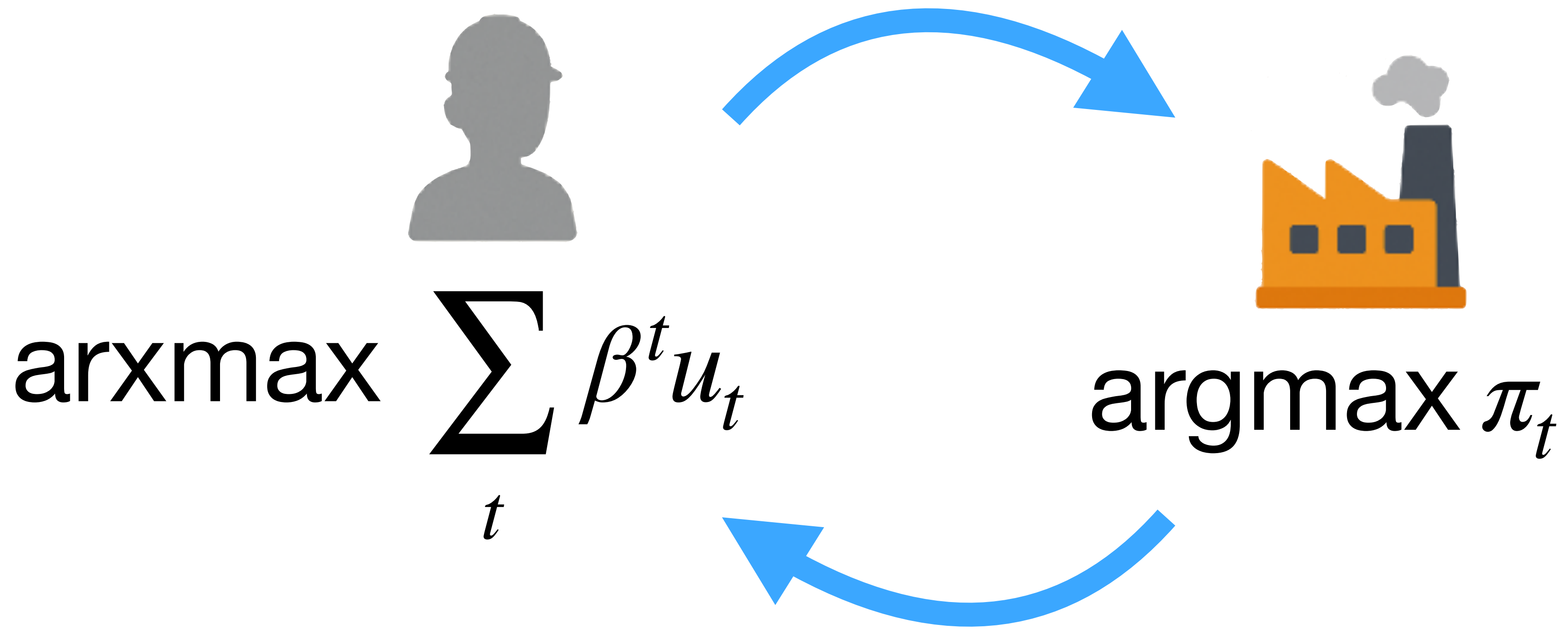
✘ Models are **hard to design and analyse…**



- A **computational framework** that represents an economy as the **outcome of interactions among heterogeneous, boundedly rational agents** whose behavior and expectations emerge from adaptive rules rather than imposed equilibrium conditions

DSGEs in a nutshell: the RBC

- ✗ Very limited or completely absent **heterogeneity**
- ✗ **Unrealistic behaviour** coming from assumptions of perfect “rationality”
- ✓ Behavioural **rules are easy to specify** in advance
- ✓ Behaviour optimally adapts to **changes in policy**
- ✓ Models are **simple to design** and analyse



$$\mathscr{L} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \Big[u(c_t, 1-\ell_t) + \lambda_t\big(Y(k_t,\ell_t) - c_t - i_t\big) + \mu_t\big((1-\delta)k_t + i_t - k_{t+1}\big)\Big]$$

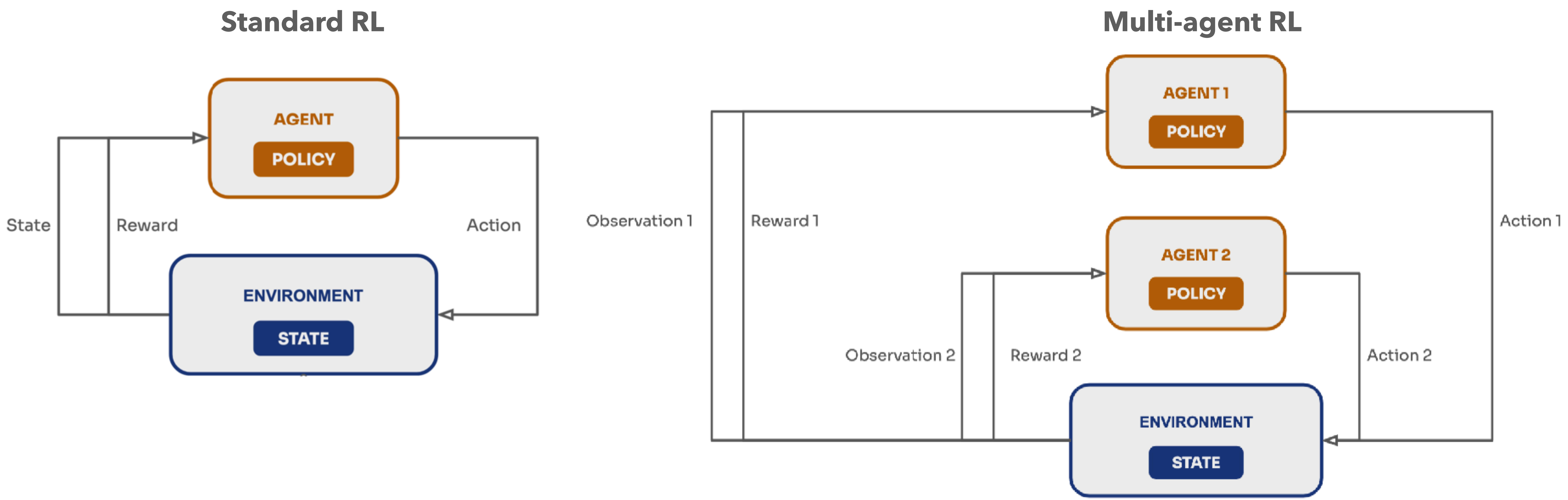
- A **framework** that represents the economy as a **general equilibrium** outcome of intertemporally **optimising representative agents** facing stochastic shocks

ABMs and DSGEs

✓ Unconstrained **heterogeneity**	✗ Very limited or completely absent **heterogeneity**…
✓ **Realistic**, boundedly rational behaviour	✗ **Unrealistic behaviour** coming from assumptions of perfect "rationality"…
✗ Behavioural **rules are difficult to specify** in advance…	✓ Behavioural **rules are easy to specify** in advance
✗ Behaviour does not necessarily adjust as a result of a **changing policy**…	✓ Behaviour optimally adapts to **changes in policy**
✗ Models are **hard to design** and analyse…	✓ Models are **simple to design** and analyse

Can we do better by leveraging modern computation?

(Multi-agent) reinforcement learning



- **Reinforcement Learning (RL):** an agent learns optimal behavior through trial and error by maximizing cumulative rewards from interactions with an environment

- **Multi-Agent Reinforcement Learning (MARL):** multiple agents learn and adapt simultaneously, each optimizing its own rewards while interacting within a shared environment

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ABMs, DSGEs, MARLs

Feature	**ABMs**	**DSGEs**	**MARLs**
Unconstrained **heterogeneiuty**	✓	✗	✓
Realistic boundedly rational behaviour	✓	✗	✓
Behaviour adapts to **chaning policy**	✗	✓	✓
Ease of **behavioural specification**	✗	✓	✓
Ease of **model development**	✗	✓	✗

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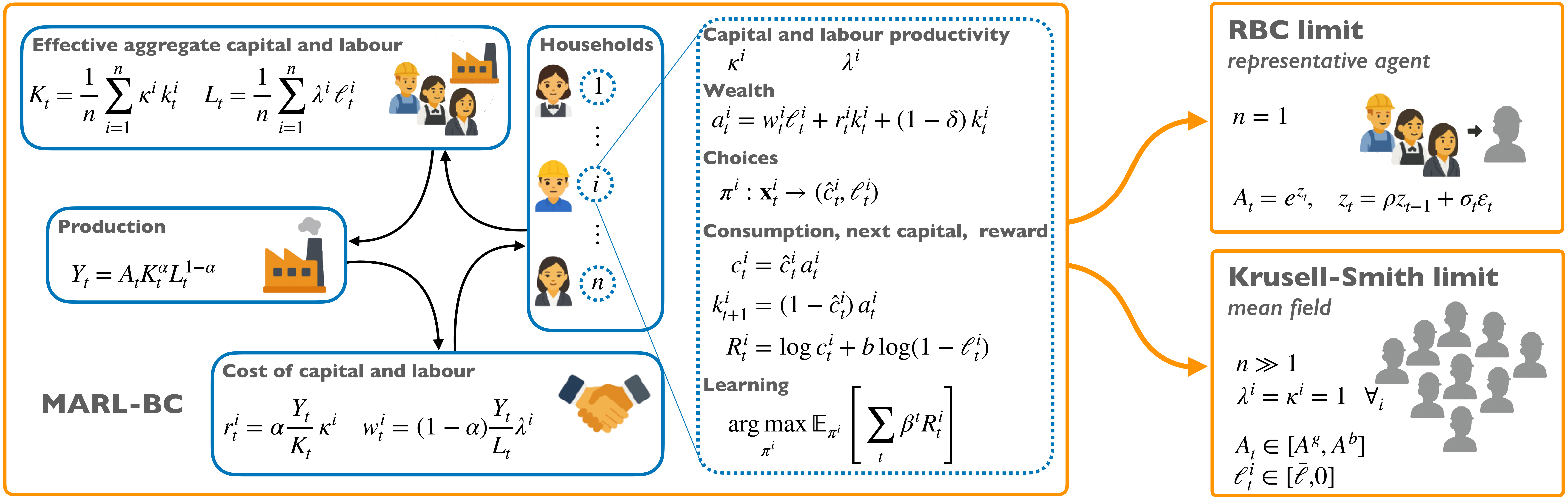
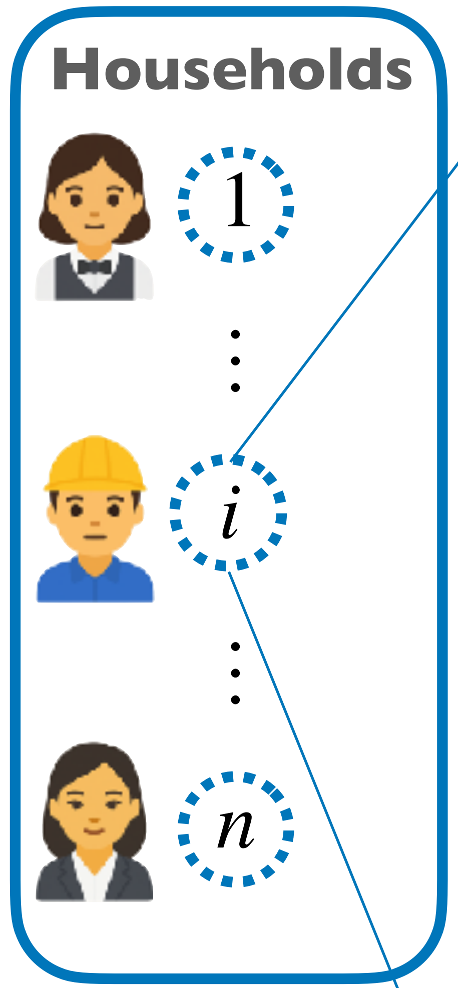
Extending ABMs and DSGEs with MARL



- The *Multi-Agent Reinforcement Learning Business Cycle* (***MARL-BC***) is a MARL-based framework that **extends the classic RBC with arbitrary heterogeneity**
- MARL-BC can recover the **classic RBC** results **using a single agent**
- MARL-BC can recover the **mean-field Krusell-Smith** using a large population of **ex-ante identical agents**

MARL-BC - learning scheme

network size: |x| - 265 - 256 - 2



Capital and labour productivity

$\kappa^i \qquad \lambda^i$

Wealth

$$a_t^i = w_t^i \ell_t^i + r_t^i k_t^i + (1-\delta)\, k_t^i$$

Choices

$$\pi^i : \mathbf{x}_t^i \to (\hat{c}_t^i, \ell_t^i)$$

Consumption, next capital, reward

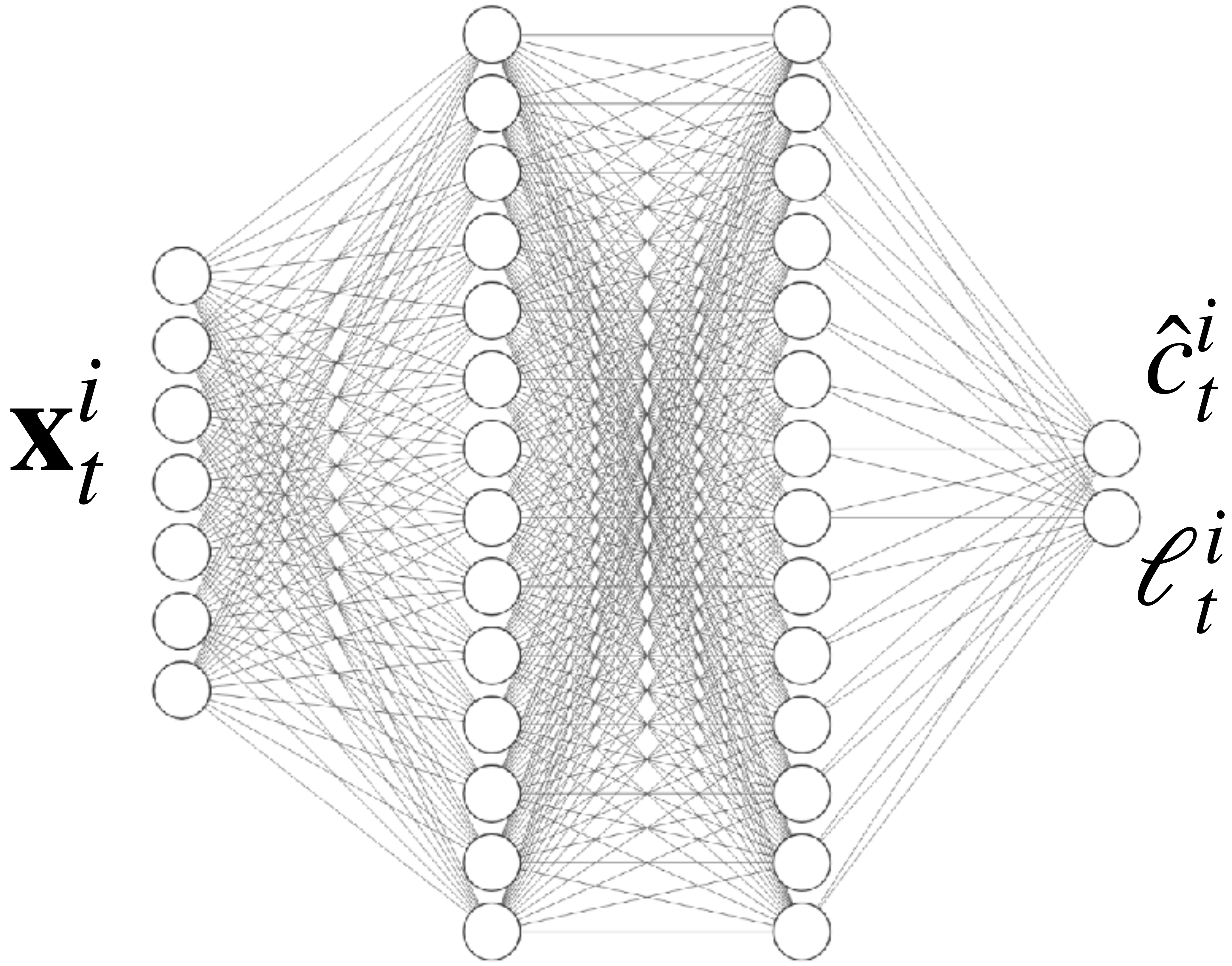
$$c_t^i = \hat{c}_t^i\, a_t^i$$

$$k_{t+1}^i = (1-\hat{c}_t^i)\, a_t^i$$

$$R_t^i = \log c_t^i + b \log(1-\ell_t^i)$$

Learning

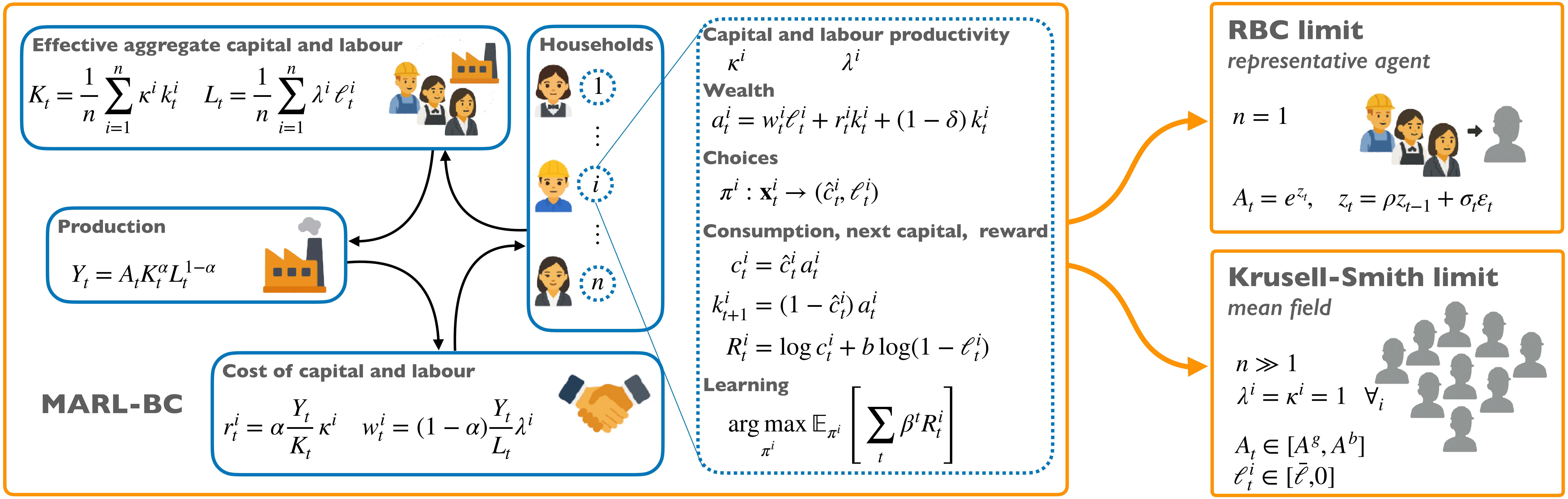
$$\arg\max_{\pi^i} \mathbb{E}_{\pi^i}\left[\sum_t \beta^t R_t^i\right]$$



$$\mathbf{x}_t^i = (k_t^i, K_t, \ell_{t-1}^i, L_{t-1}, A_t, \kappa^i, \lambda^i, i)$$

- **Parameters are shared** among all agents for faster convergence (a rudimentary model of "social learning")
- **Heterogeneity** in the behaviour of agents is still guaranteed **by different agent observations**

MARL-BC - Results



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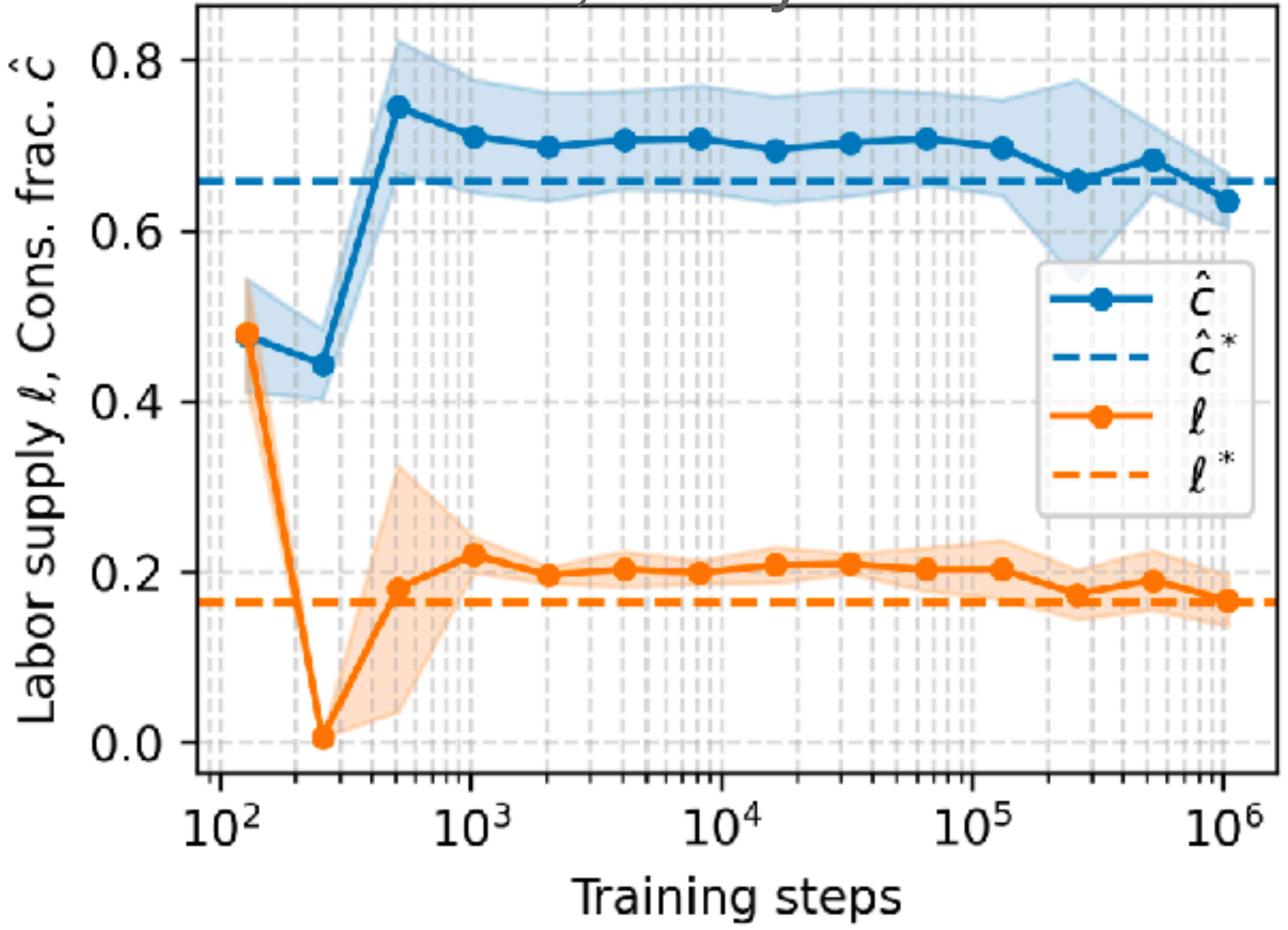
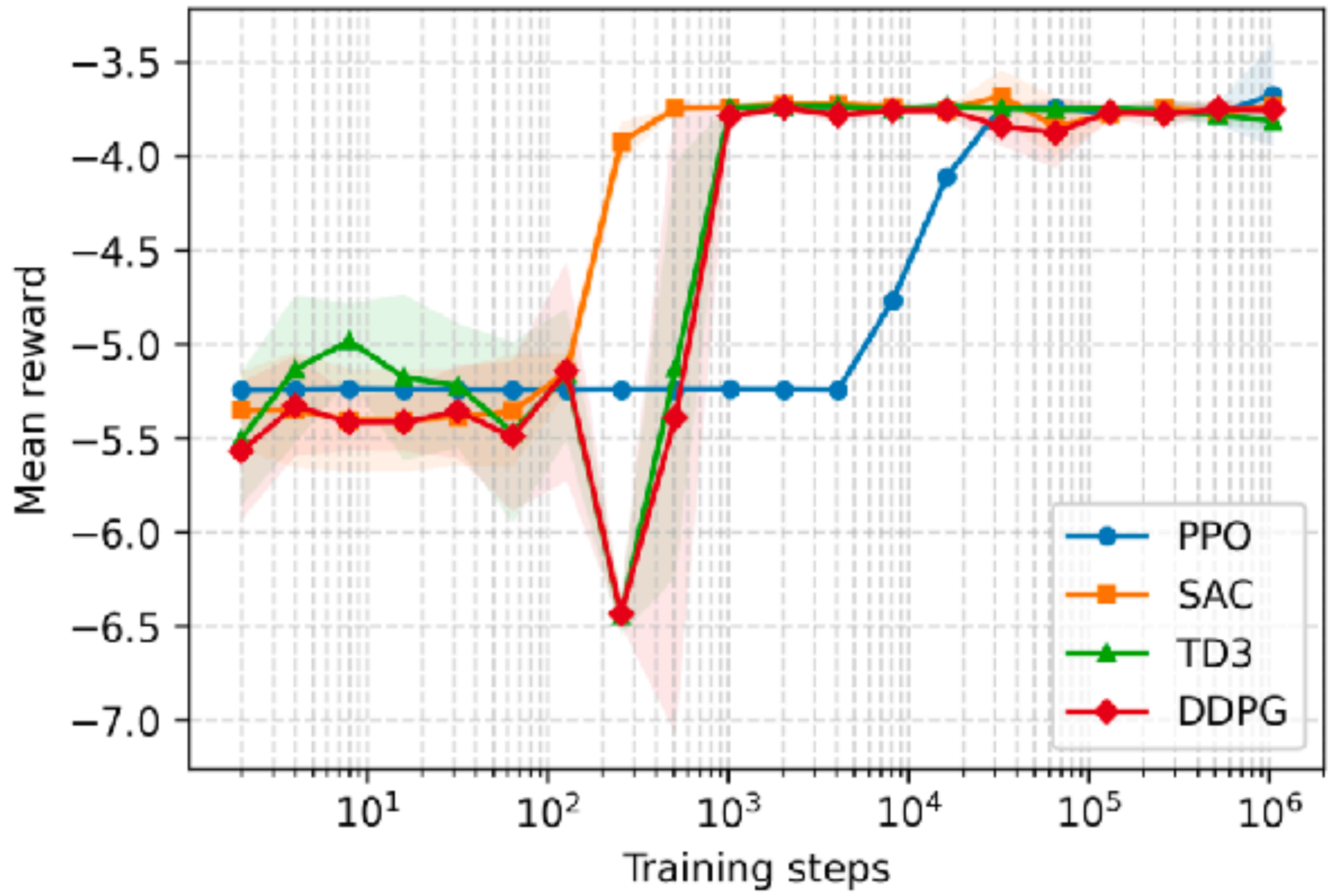
MARL-BC can recover the **classic RBC**

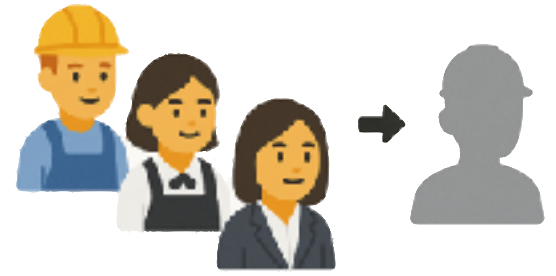
$\delta = 1$, analytical sol.





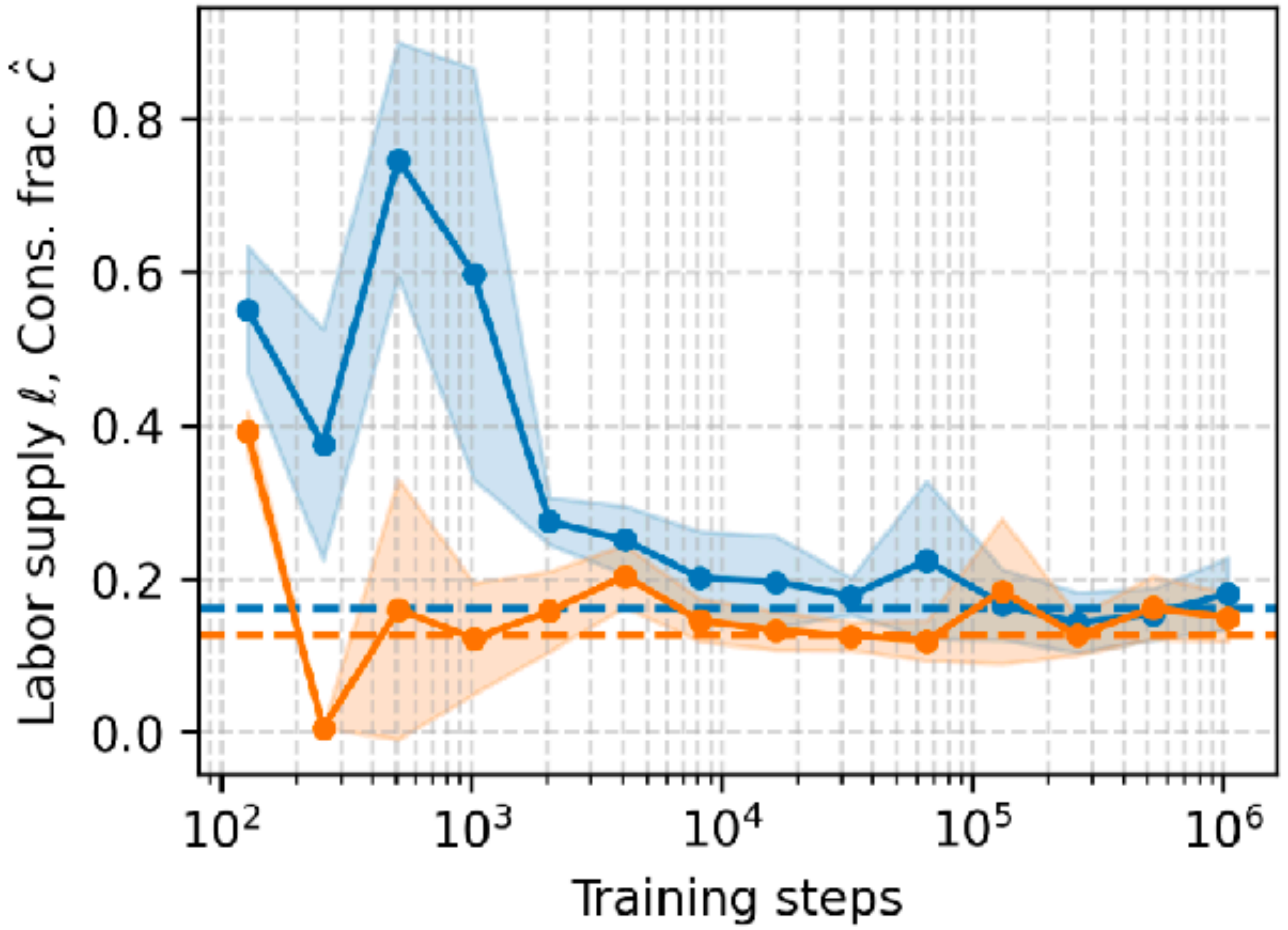
RBC limit
representative agent

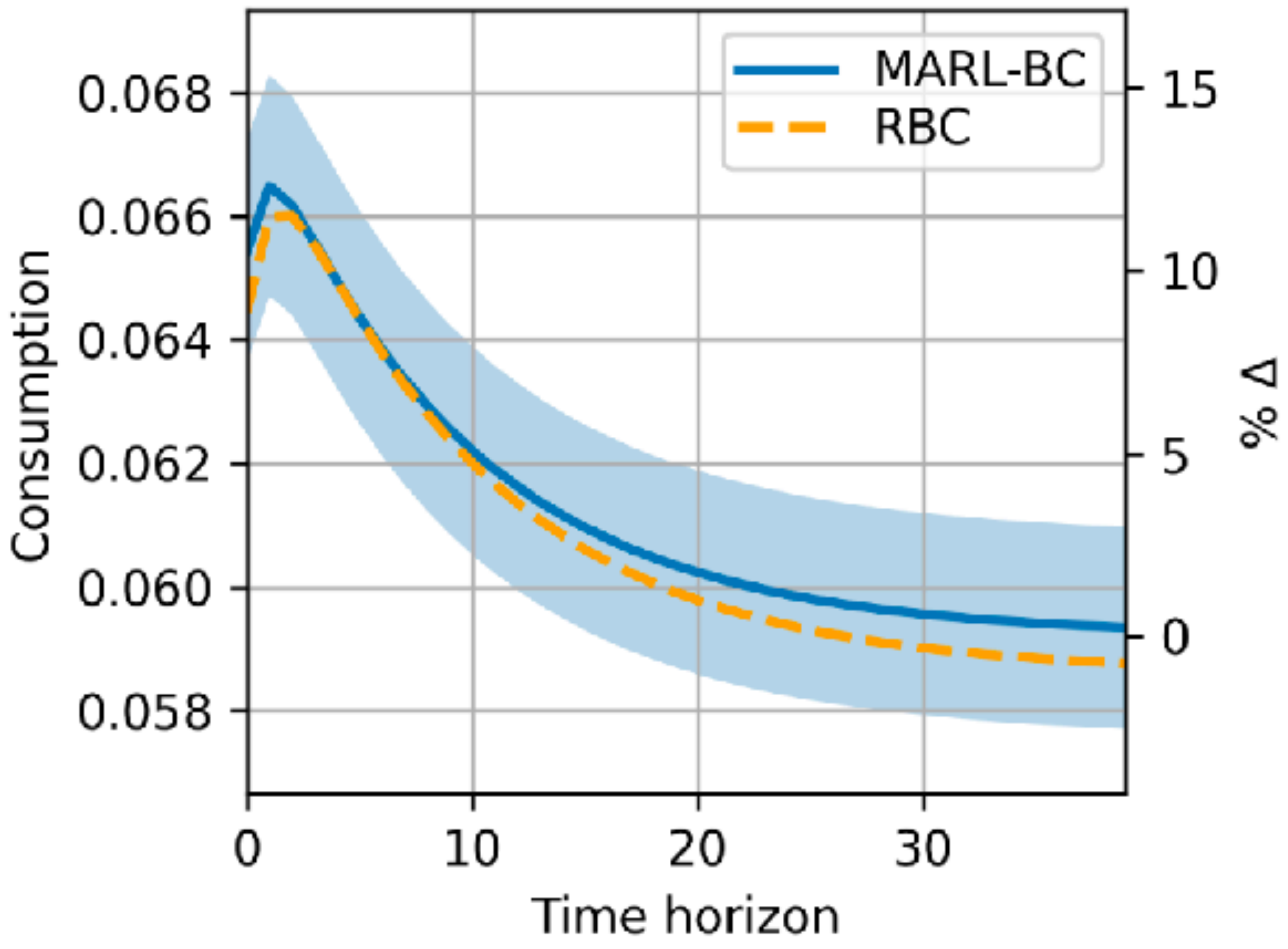
$n = 1$



$A_t = e^{z_t}, \quad z_t = \rho z_{t-1} + \sigma_t \varepsilon_t$

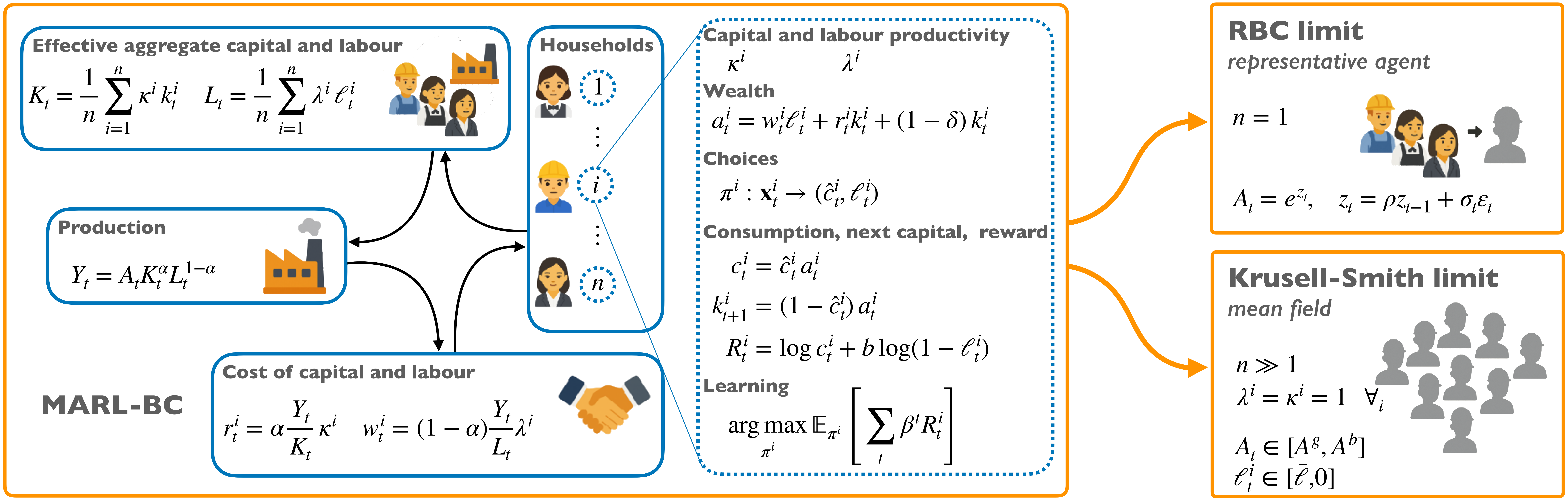
$\delta = 0.025$, Dynare sol.



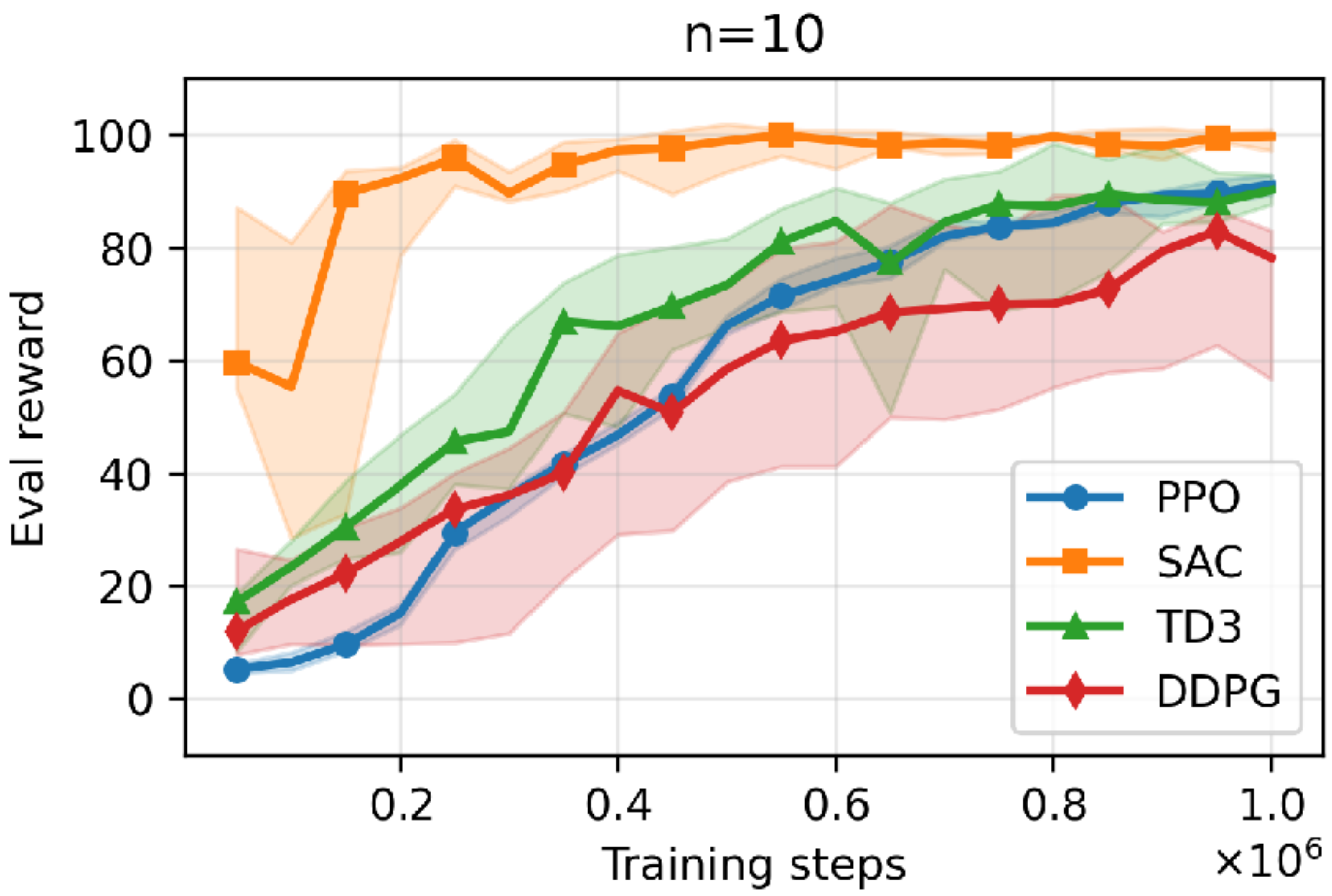


- It is solved by **state-of-the-art RL** schemes
- It reproduces **analytical and Dynare** solutions
- It also reproduces **dynamic behaviour** (IRFs)

Connecting ABMs and DSGEs with MARL



MARL-BC "equilibrium" convergence (KS)



Krusell-Smith limit

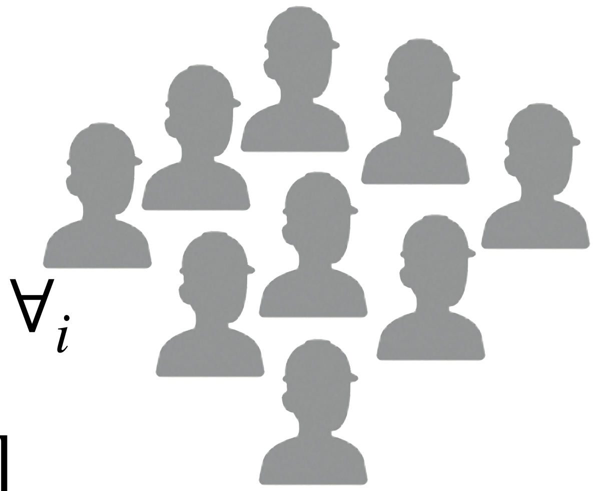
mean field

$n \gg 1$

$\lambda^i = \kappa^i = 1 \quad \forall_i$

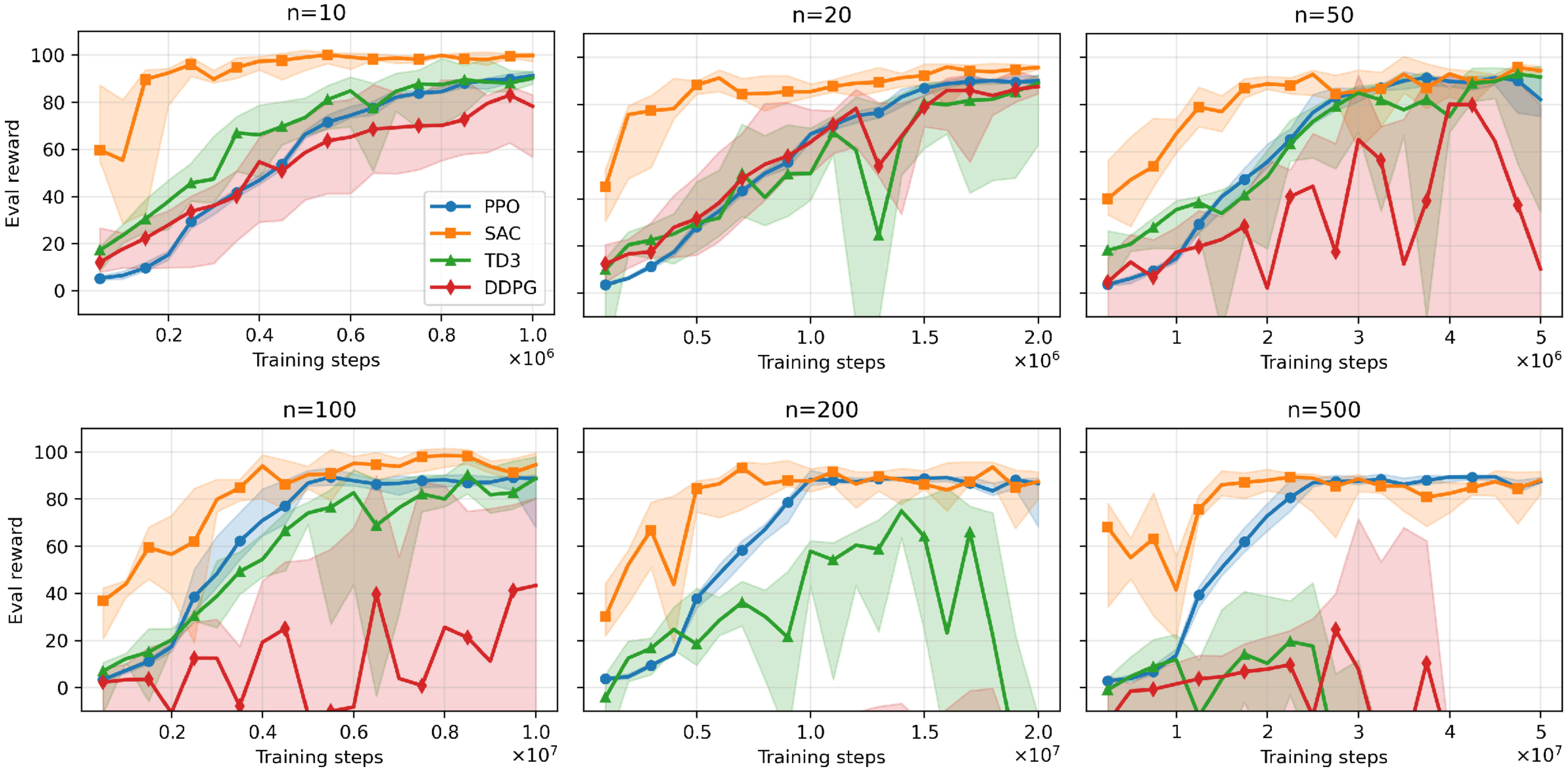
$A_t \in [A^g, A^b]$

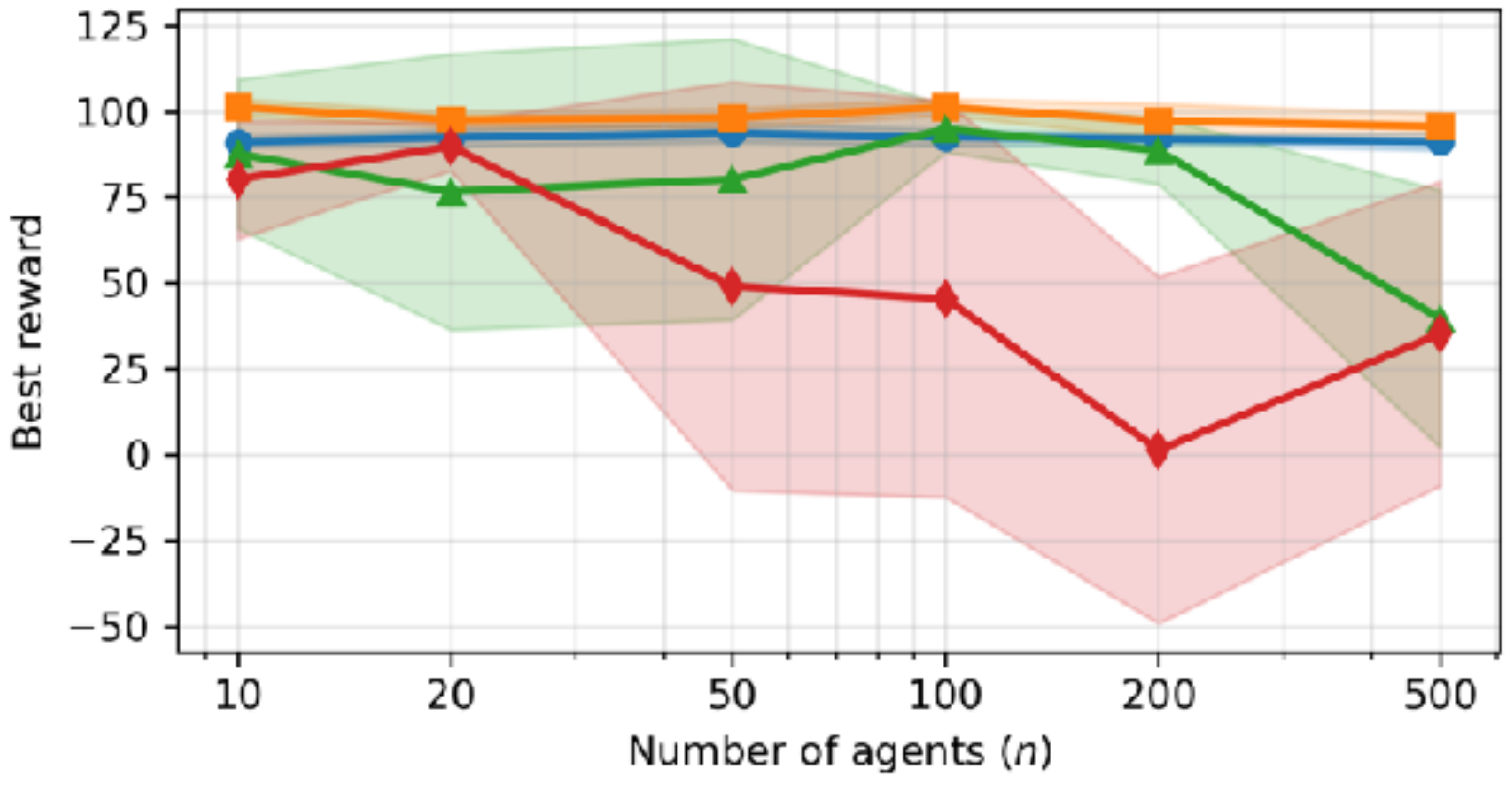
$\ell_t^i \in [\bar{\ell}, 0]$



- **Soft Actor-Critic (SAC)** reliably achieves high rewards

MARL-BC "equilibrium" convergence (KS)





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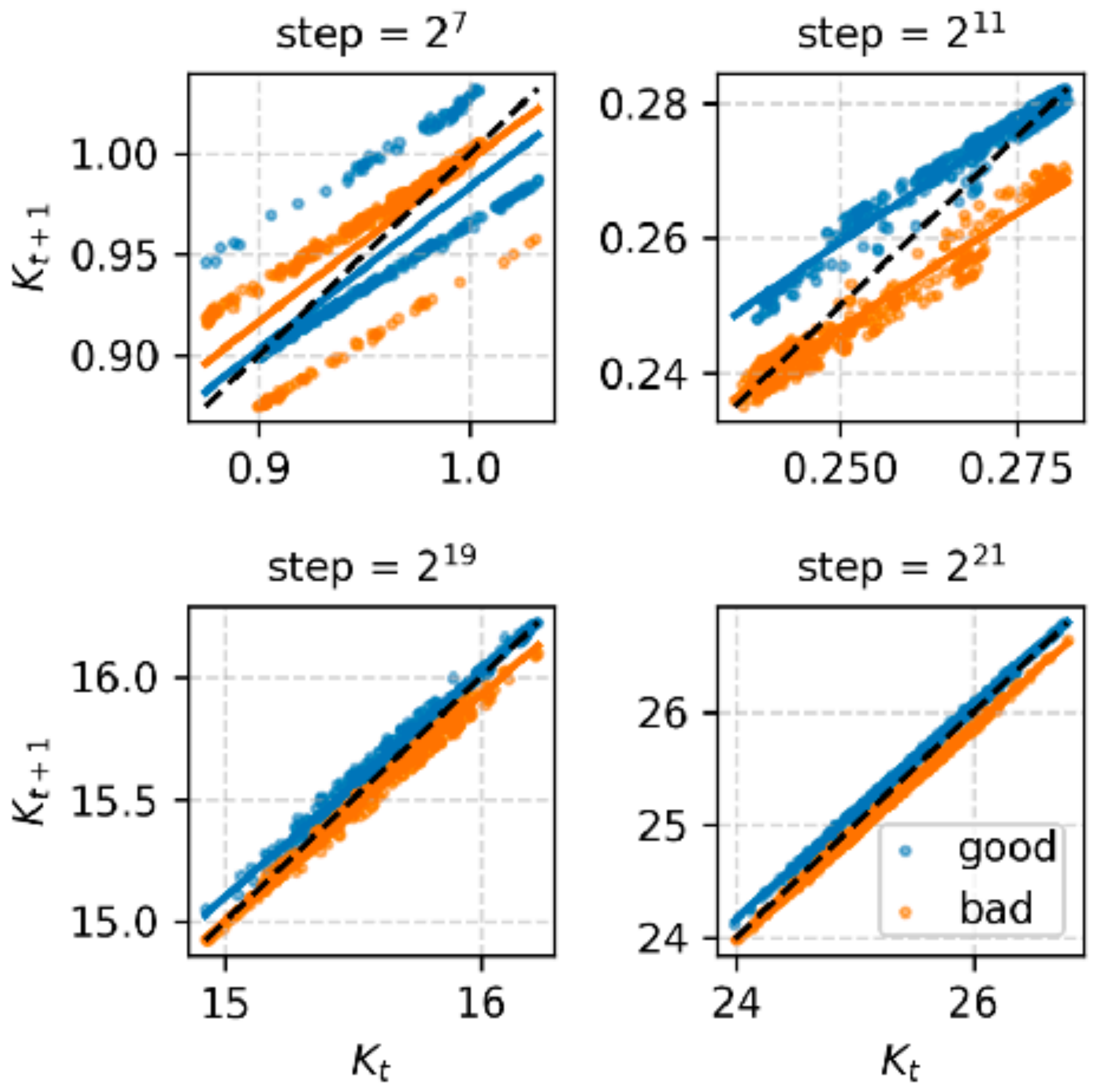
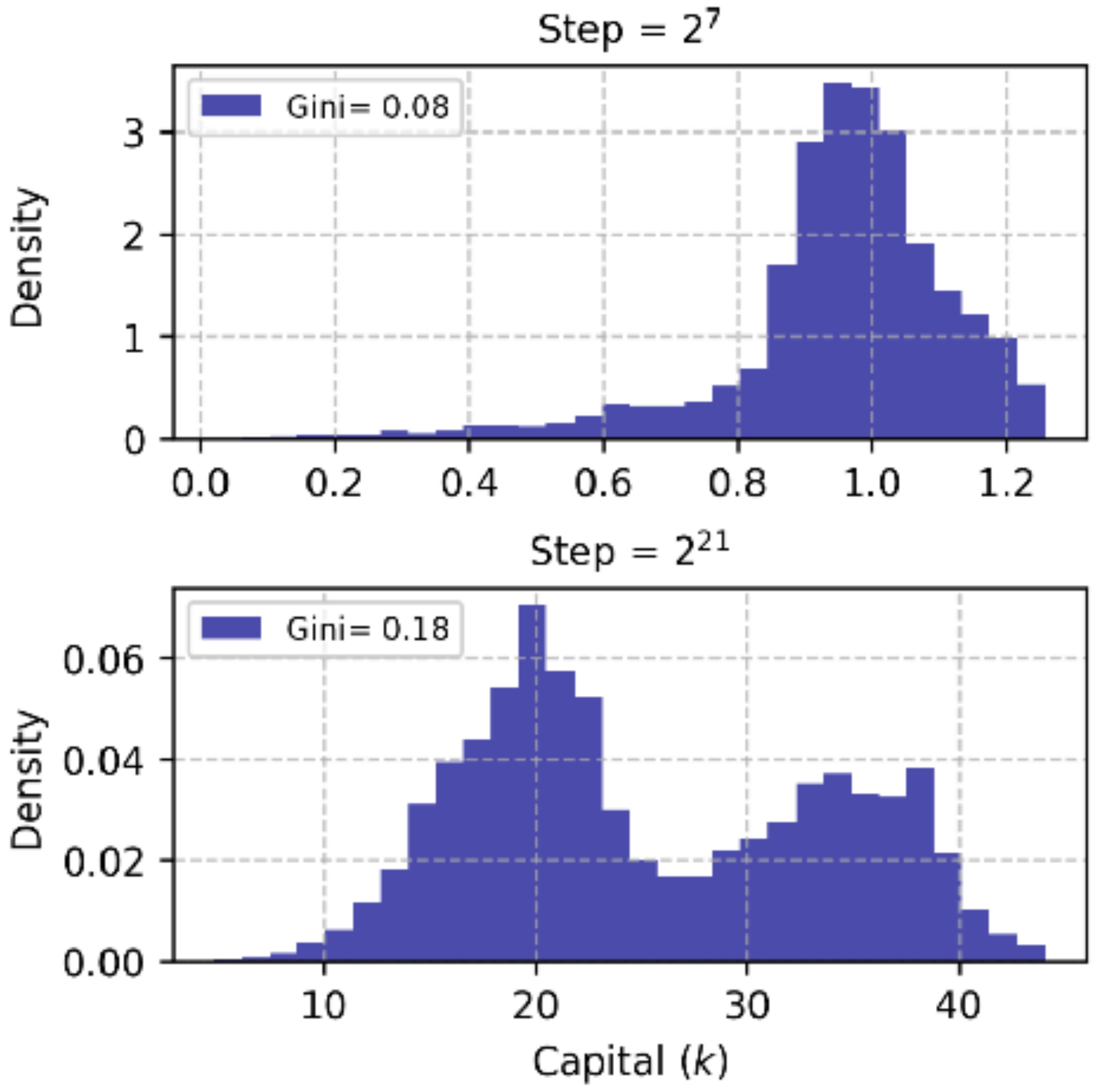
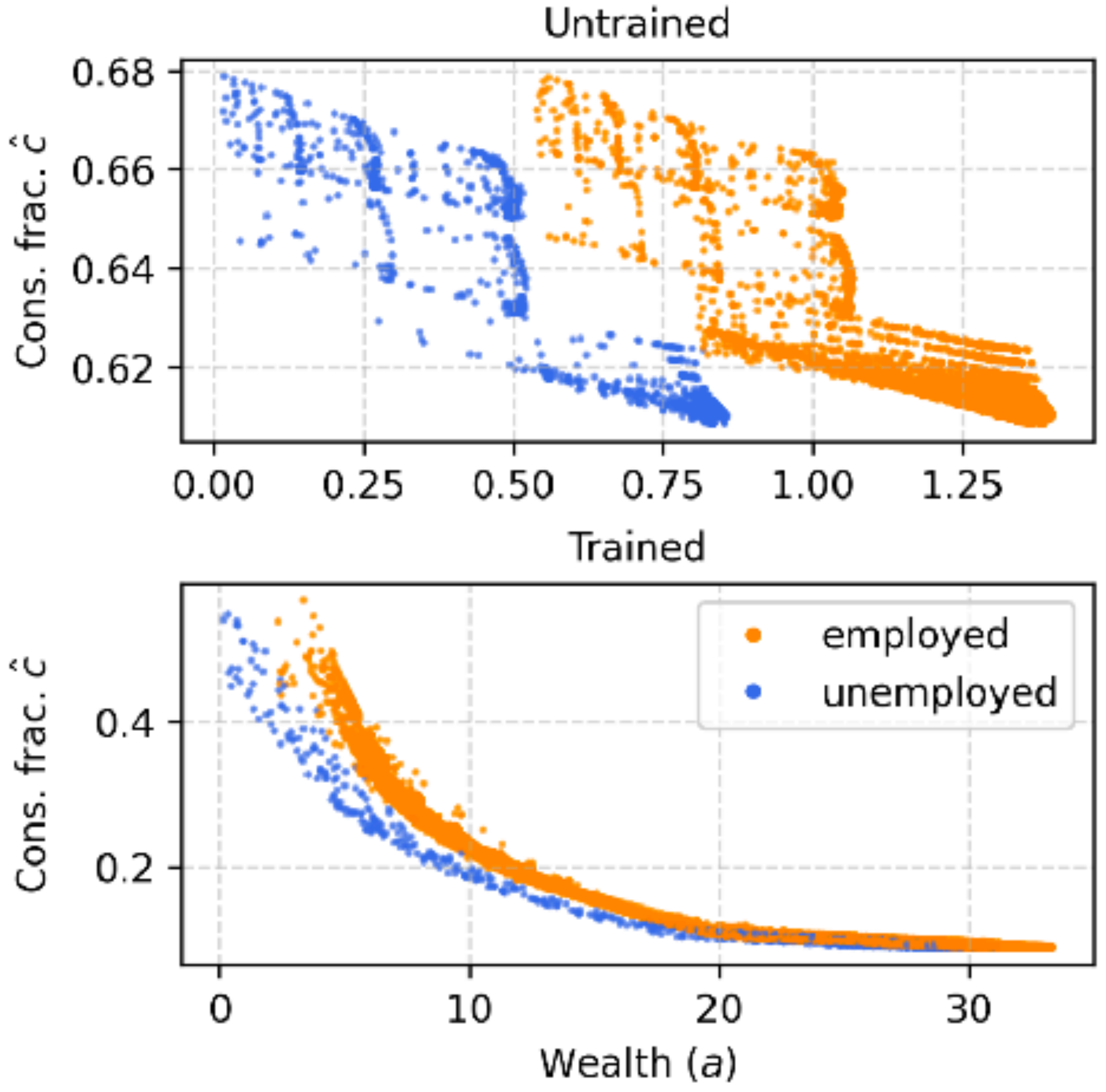
- **Soft Actor-Critic (SAC)** reliably achieves high rewards
- **Best rewards are stable** for large populations

MARL-BC can recover the **mean-field Krusell-Smith**







- The **'law of motion' for aggregate capital** becomes linear with training, as just *postulated* in the original paper
- The **Gini index** increases to 0.18 with training, a value relatively close to the 0.25 value from the original paper
- The **marginal propensity to consume** becomes flat for large values of wealth and increase rapidly for low values of wealth

Krusell-Smith limit

mean field

$n \gg 1$

$\lambda^i = \kappa^i = 1 \quad \forall_i$

$A_t \in [A^g, A^b]$

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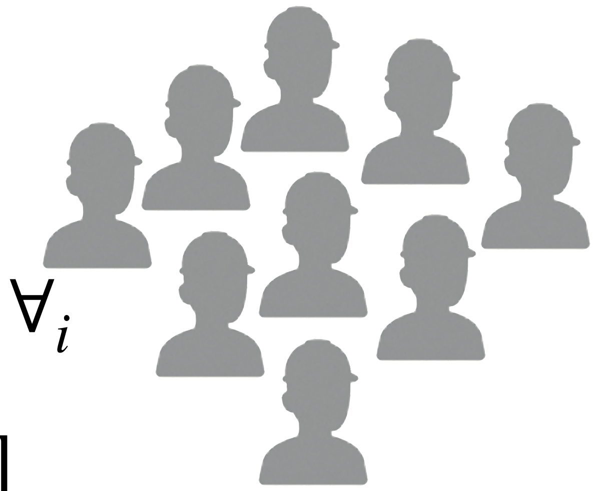
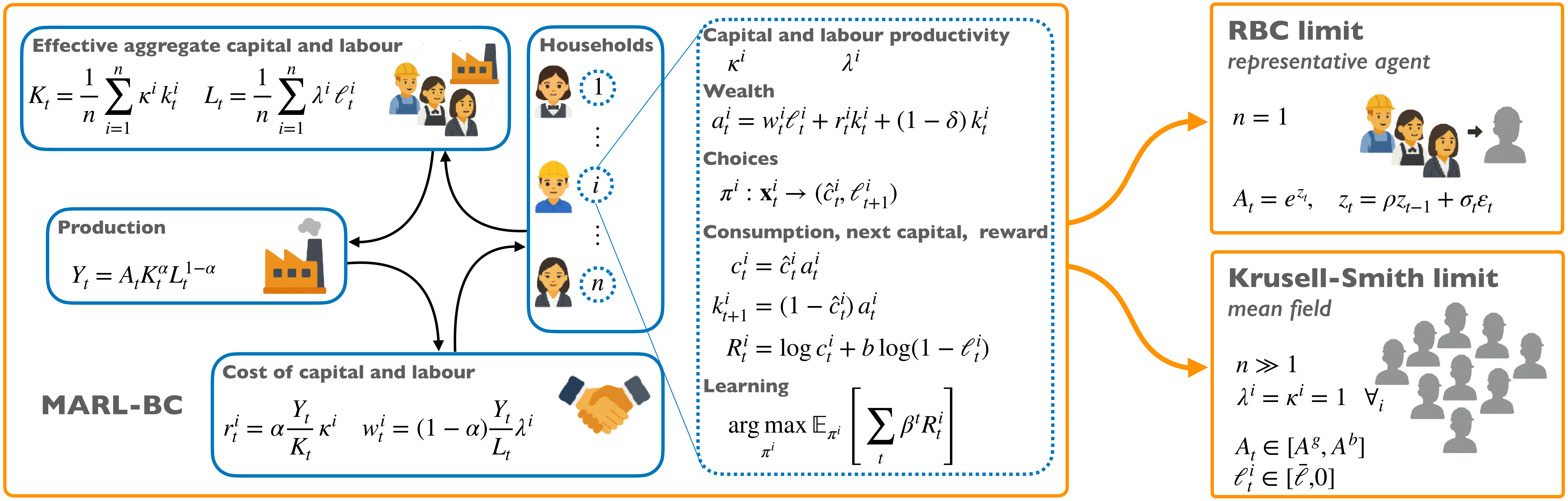
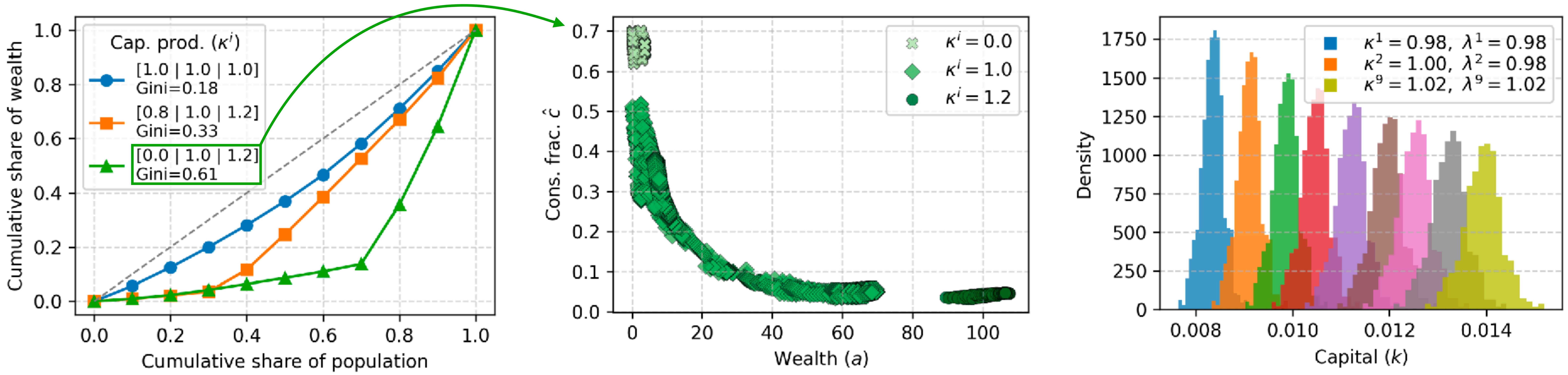


Connecting ABMs and DSGEs with MARL



MARL-BC allows for greater agent heterogeneity



- Extending the **KS with groups of heterogeneous returns** (capital prods.) allows for the modelling of more unequal economies

- Consumers with low capital returns **endogenously learn a 'hand to mouth' policy**, often encoded in GE models as a heuristic

- Extending the RBC with **heterogeneous returns and wages** produces different wealths and consumption/labour choices

MARL-BC - general case $n \gg 1$ $\lambda^i \neq \kappa^i \quad \forall_i$

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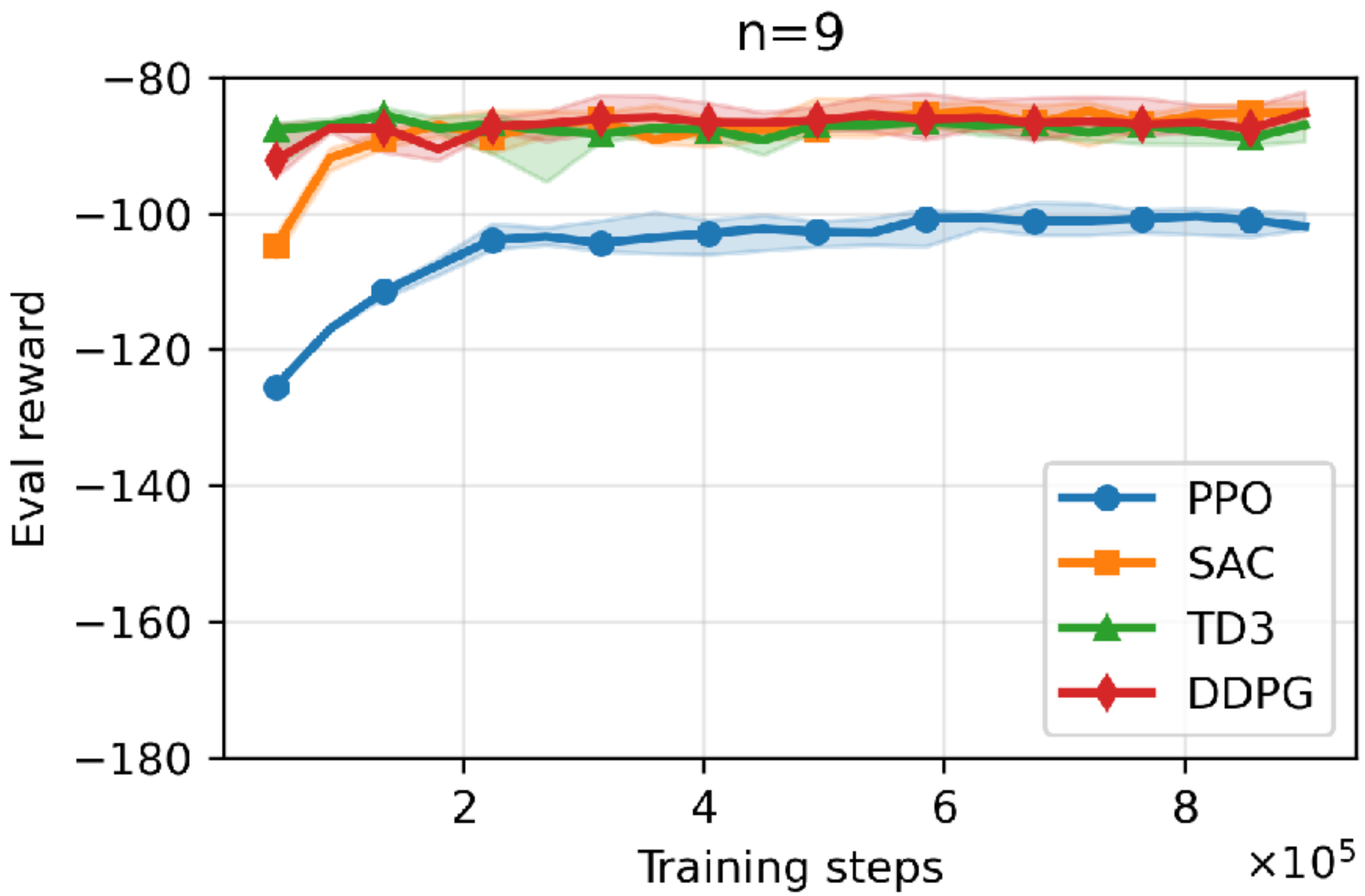
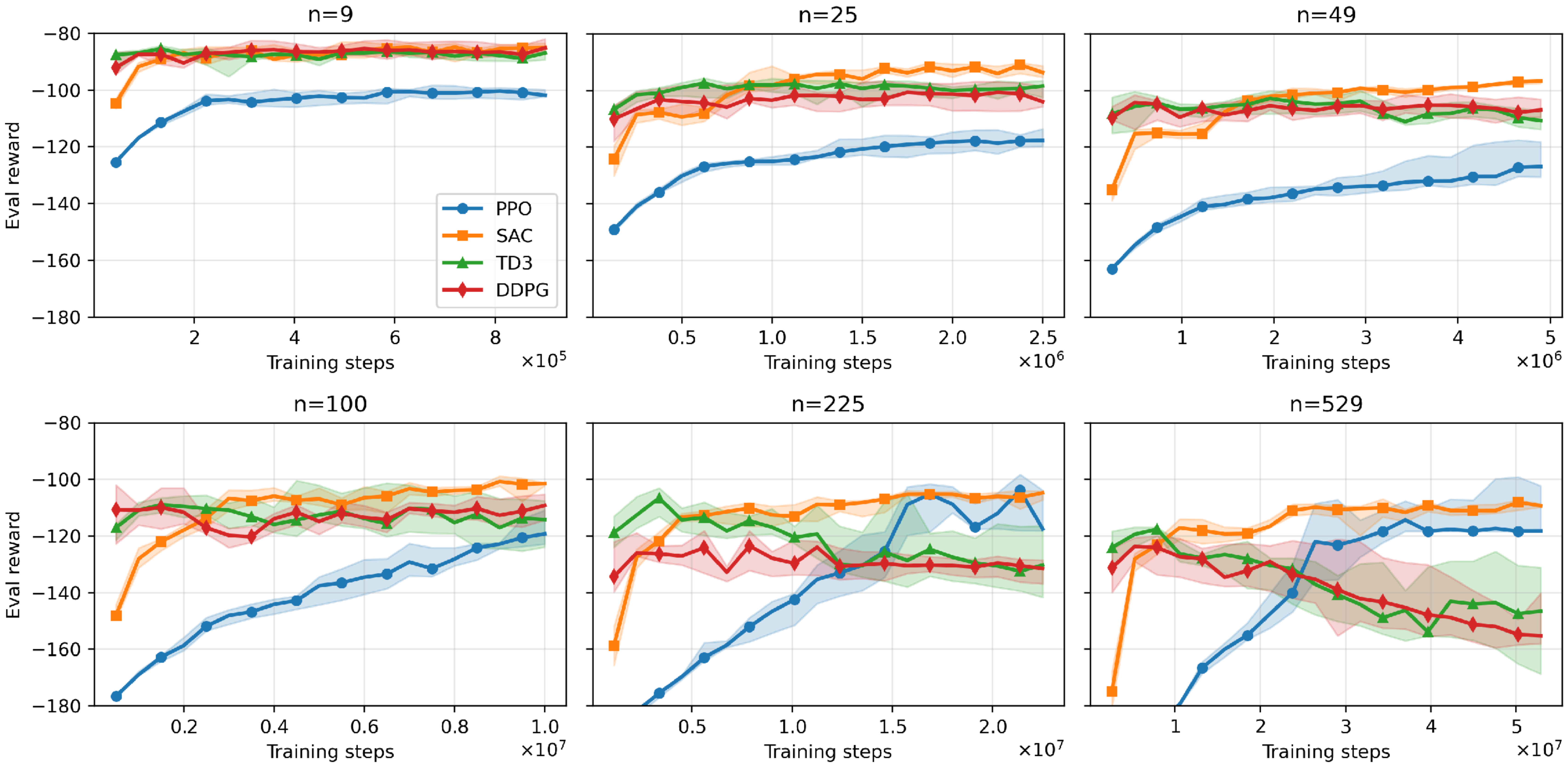
MARL-BC "equilibrium" convergence (RBC w. grid-het)

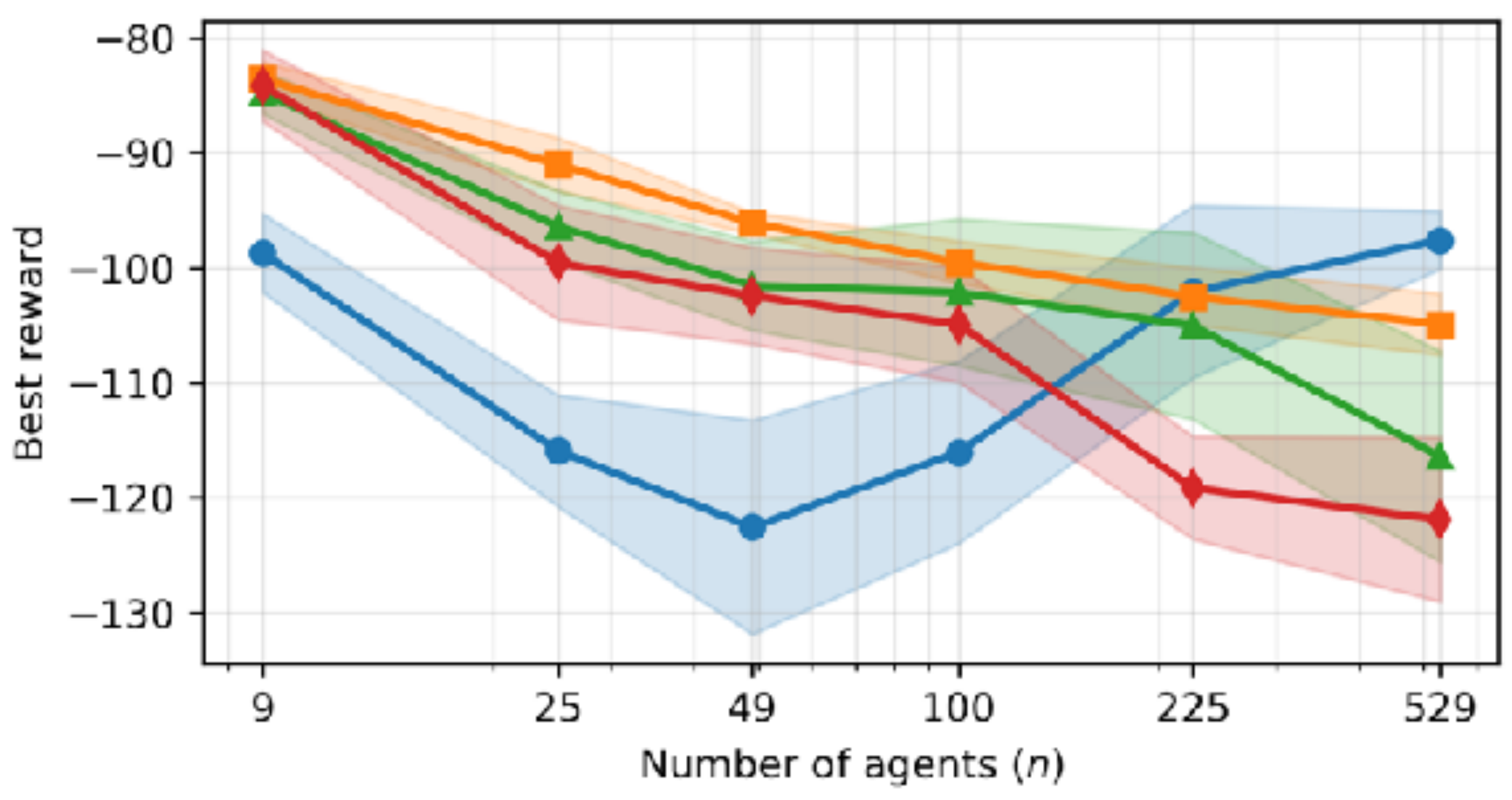


MARL-BC - general case $n \gg 1$ $\lambda^i \neq \kappa^i \quad \forall_i$

- **Soft Actor-Critic (SAC)** reliably achieves high rewards

MARL-BC “equilibrium” convergence (RBC w. grid-het)





MARL-BC - general case $n \gg 1$ $\lambda^i \neq \kappa^i \quad \forall_i$

- **Soft Actor-Critic (SAC)** reliably achieves high rewards
- **Best rewards are stable** for large enough populations

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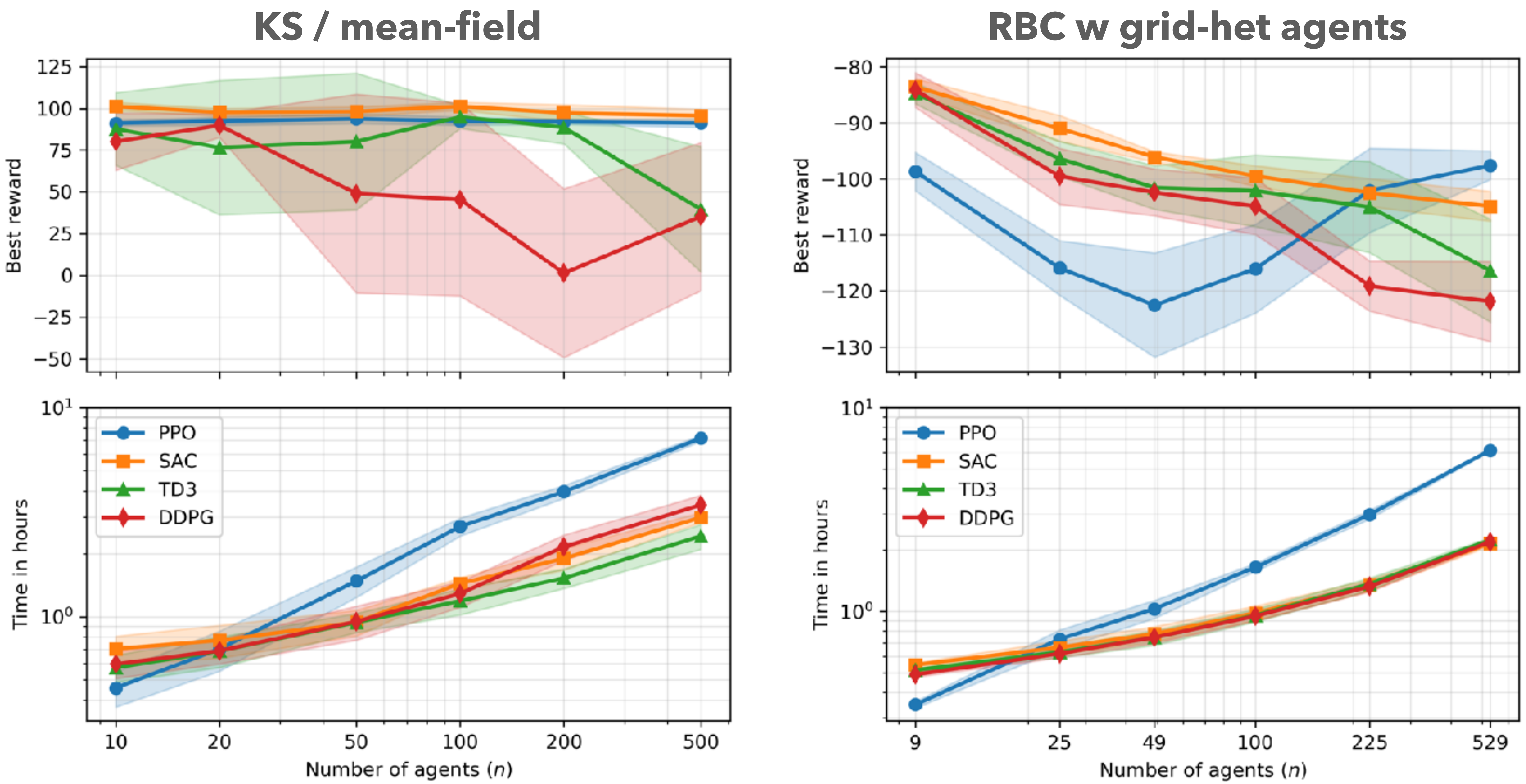
MARL-BC scalabilty to large sizes



- Best rewards are stable **for lage enough populations** in both cases
- Training takes around **one hour on a single CPU**!

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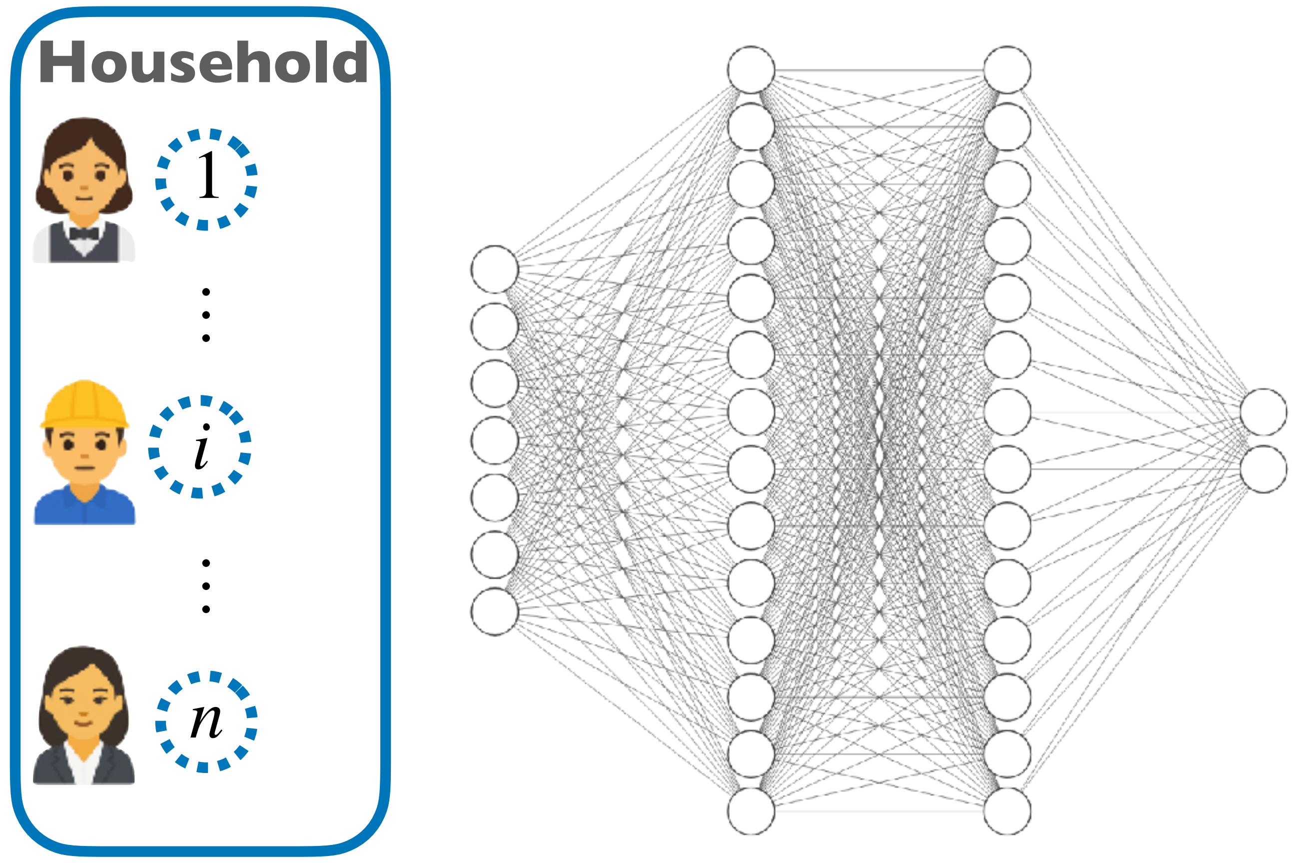
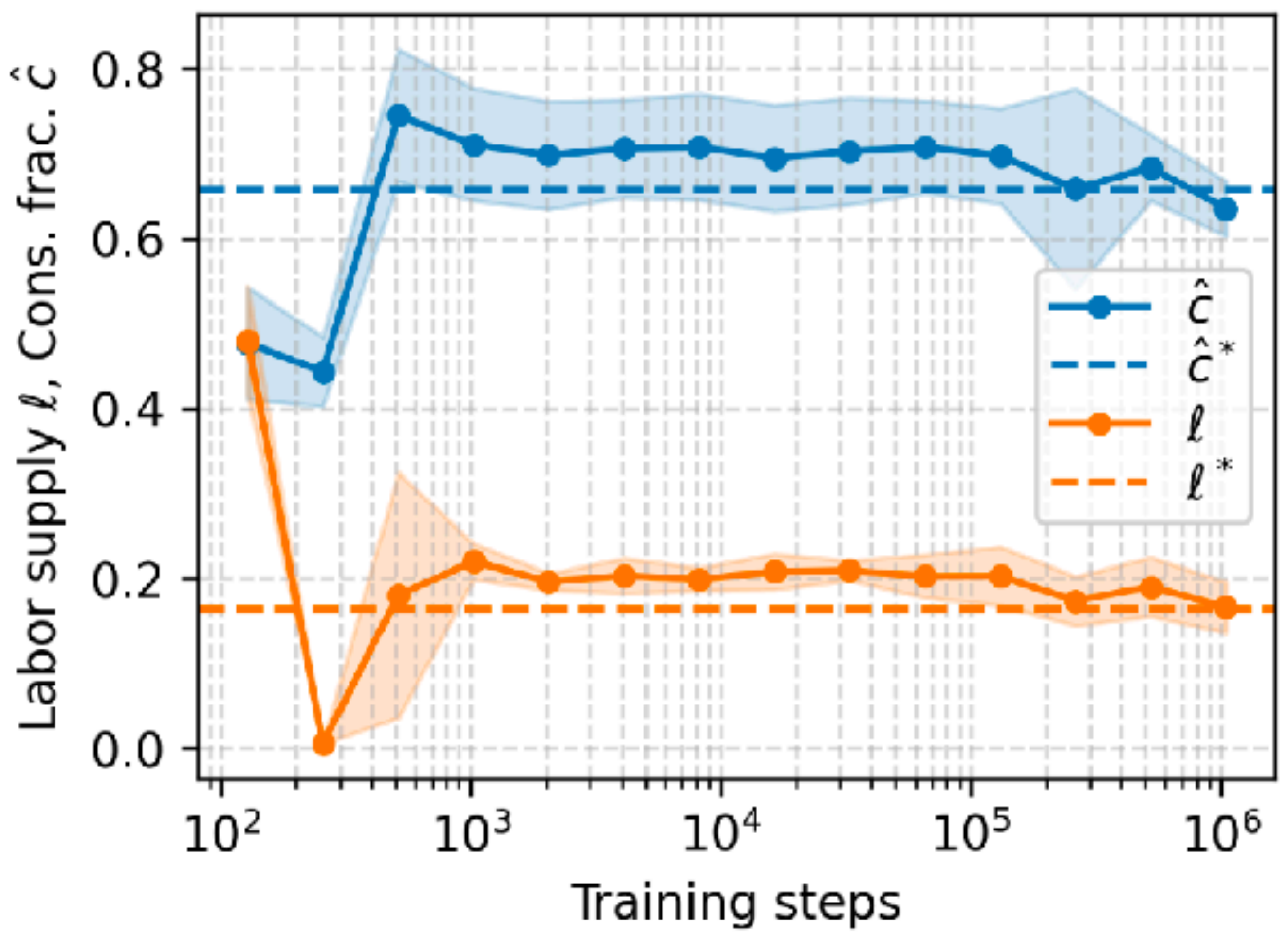
Summary of contributions

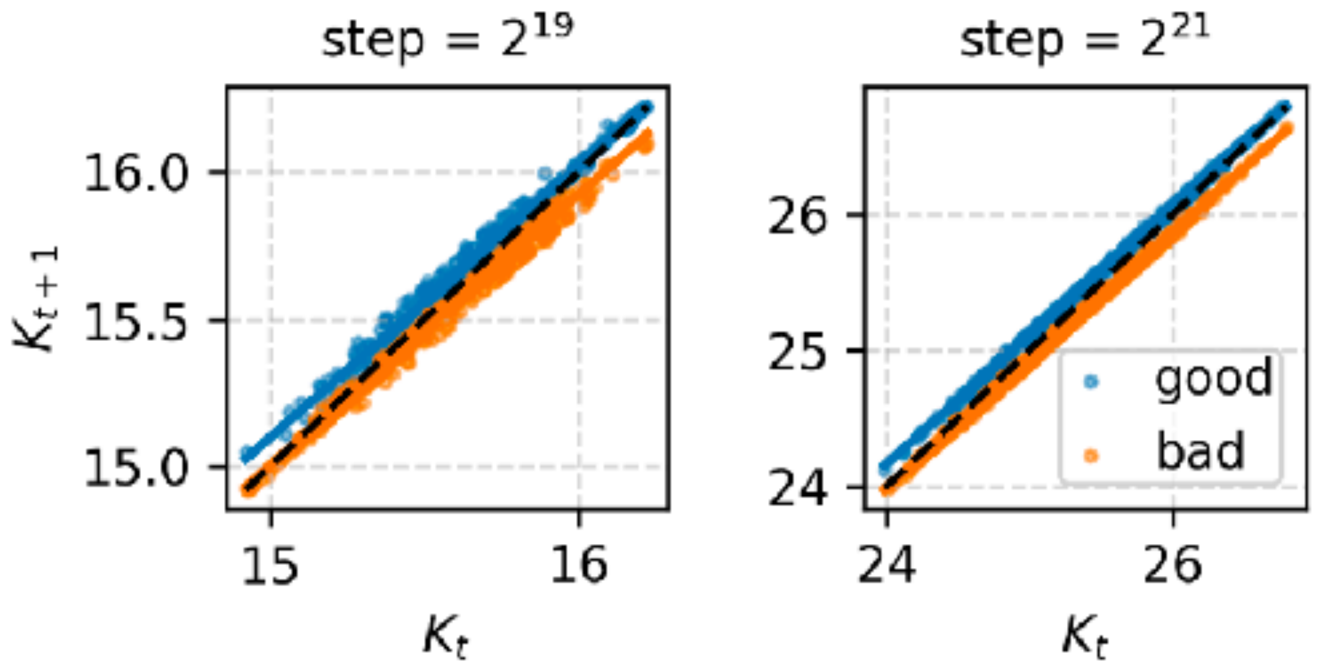


- Developed **MARL-BC** to integrate deep multi-agent reinforcement learning (MARL) with real business cycle (RBC) models

- Demonstrated that MARL-BC can:
  - Replicate **textbook RBCs** under a representative agent setting
  - Recover results of the **mean-field Krusell–Smith** models
  - Accommodate a much **more varied agent heterogeneity**



- Traced a direction for a **synthesis of DSGE and ABM approaches**, fostering dialogue between economists (DSGE/ABM) and computer scientists working on reinforcement learning



A. Gabriele, A. Glielmo, M. Taboga,
Heterogeneous RBCs via Deep Multi-Agent Reinforcement Learning,
AAMAS (2026)

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A baby-step towards realistic models?

Analytically solvable

A_t

external shocks

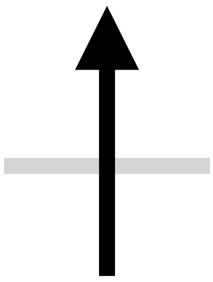
ℓ_i, k_i

representative agent

h_i

external field

$\sigma_i \in \{-1, +1\}$



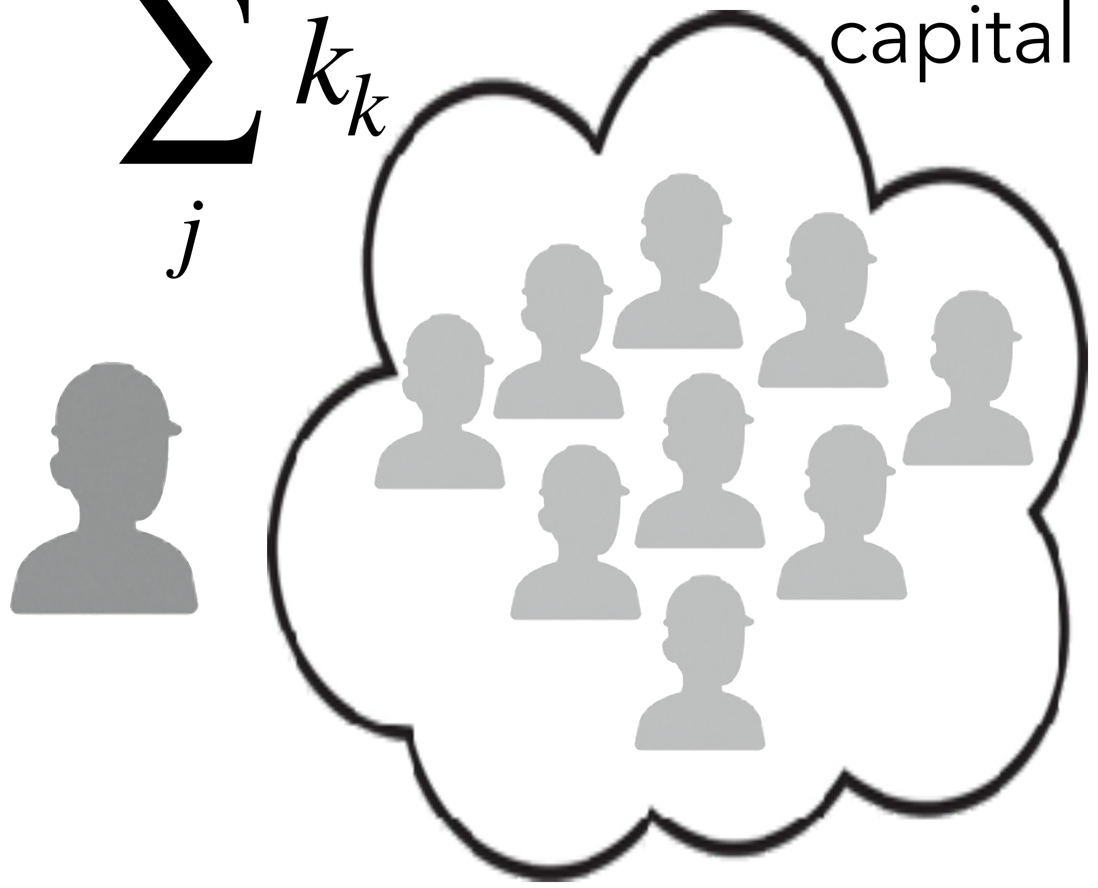
single spin

"Ising" models

Mean-field solvable

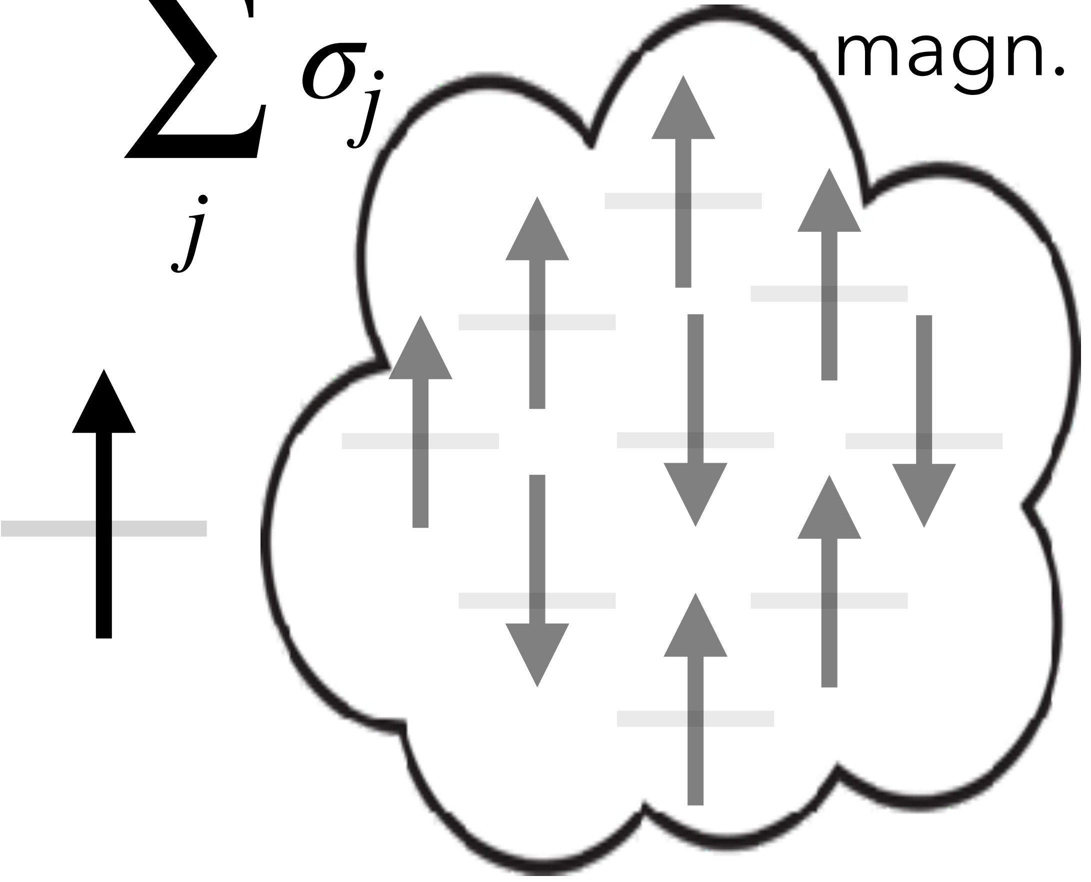
$\sum_j k_k$

Aggregate capital



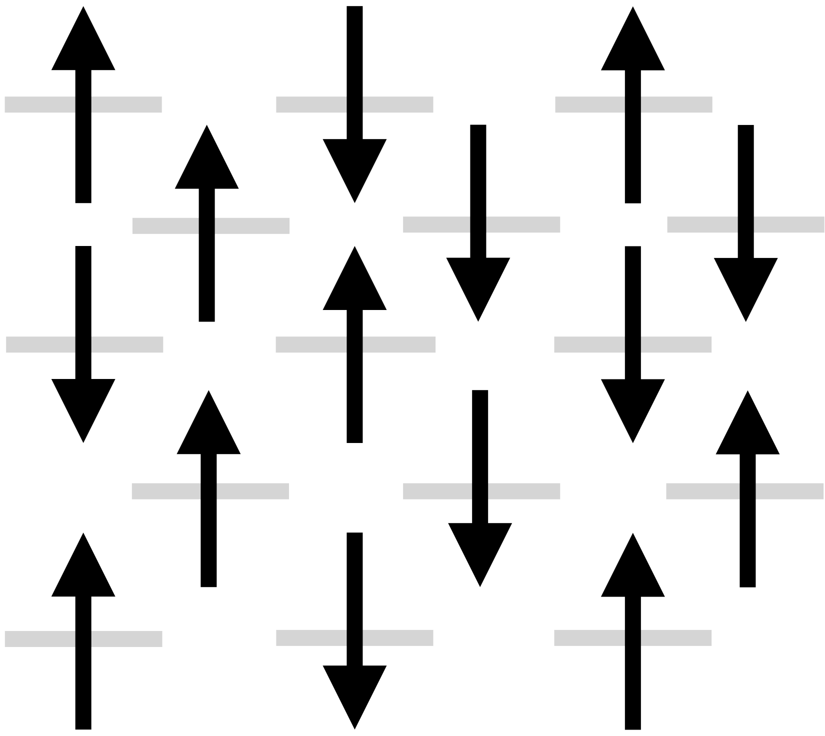
$\sum_j \sigma_j$

Average magn.



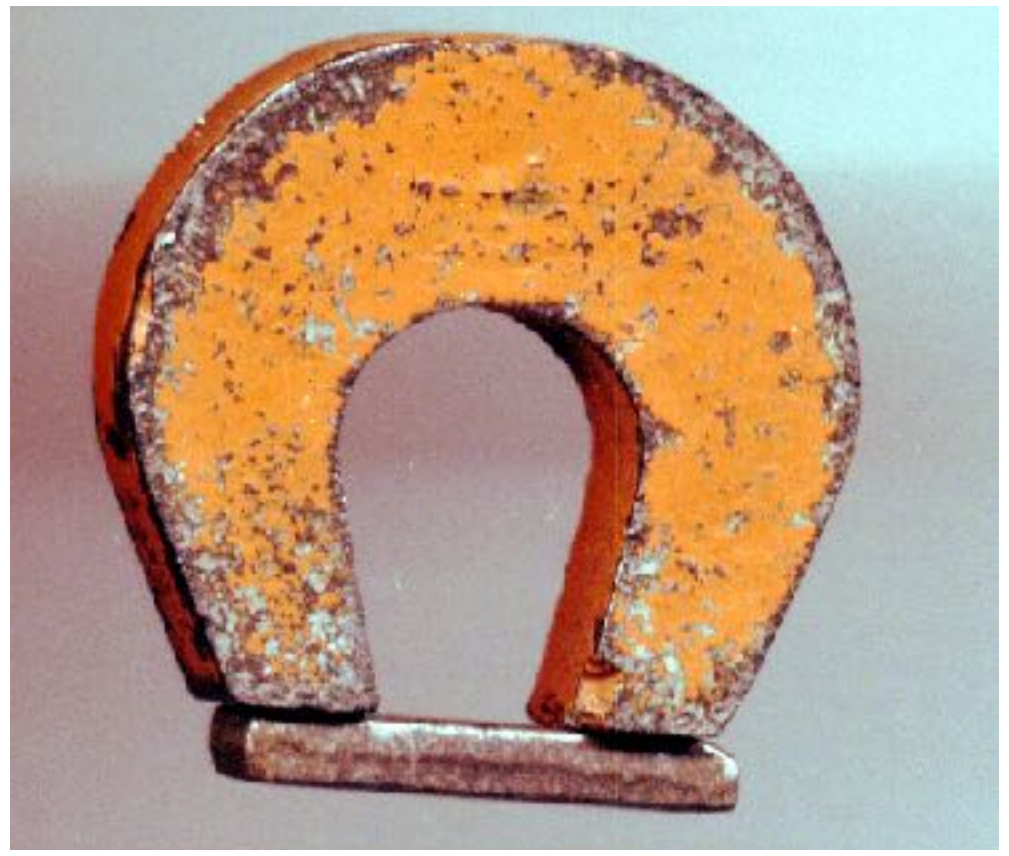
Simulation-based

MARL



Monte Carlo

Reality



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Future work?

Practical

- Develop a vectorised **GPU version of MARL-BC** to enable faster training (easily a factor of 100), improved scalability and prototyping
- Publish this work in an **economics journal** to foster dialogue between macroeconomics and reinforcement learning communities
- Introduce **MARL-NeK** — an extension embedding New Keynesian (NK) and Heterogeneous Agent New Keynesian (HANK) features within the MARL paradigm

Conceptual

- How and why MARL agents **converge to equilibrium outcomes**, and under which precise (theoretical?) conditions equilibrium is achieved?
- Are there **problem in economics** which are hard to solve with traditional methods and that are approachable with MARL-BC?

⌛ Q&A session